

Internet Appendix for “Business News and Business Cycles”

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ABSTRACT

This Internet Appendix provides additional details about our methods, as well as additional tables and figures supporting the main text.

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IA.A. Constructing the *WSJ* Document-Term Matrix

We conduct data processing steps in the following order:

1. Remove all articles prior to January 1984 and after June 2017 (data purchased at the beginning of July 2017).
2. Replace all non-alphabetical characters with an empty string and set the remaining characters to lower-case.
3. Parse article text into a white-space-separated word list retaining the article's word ordering. Exclude single-letter words.
4. Exclude articles with page-citation tags corresponding to any sections other than A, B, C, or missing.
5. Exclude articles corresponding to weekends.
6. Exclude articles with subject tags associated with obviously non-economic content such as sports. List of exclusions available from here:
<https://www.dropbox.com/sh/jvtvdpo56jvi5o3/AAAcAWVLtqrCnc-tAgdV0zaha?dl=0>.
7. Exclude articles with the certain headline patterns (such as those associated with data tables or those corresponding to regular sports, leisure, or books columns). List of exclusions available from here:
<https://www.dropbox.com/sh/jvtvdpo56jvi5o3/AAAcAWVLtqrCnc-tAgdV0zaha?dl=0>.

8. Concatenate articles with the same accession-number as these are chained articles.
9. Exclude articles with less than 100 words.
10. Remove common “stop” words and URL-based terms. List of exclusions is standard but available from here:
<https://www.dropbox.com/sh/jvtvdpo56jvi5o3/AAAcAWVLtqrCnc-tAgdV0zaha?dl=0>.
11. Lastly, we conduct light lemmatizing of derivative words. The following rules are applied in the order given, where 'x' is a candidate term. In each case, the stemming is only applied if the multiple terms reduce to the same stem.
 - (a) Replace trailing “ses” with “ss”
 - (b) Replace trailing “ies” with “y”
 - (c) Remove trailing “s”
 - (d) Remove trailing “ly”
 - (e) Remove trailing “ed.” Replace remaining trailing “ed” with “e”
 - (f) Replace trailing “ing” with “e”. For remaining trailing “ing” that follow a pair of identical consonants, remove “ing” and one consonant. Remove remaining trailing “ing”
 - (g) Remove words with less than 3 letters.
12. From the resulting uni-grams, generate the set of bi-grams as all pairs of (ordered) adjacent uni-grams.

13. Exclude terms (both uni-grams and bi-grams) appearing in less than 0.1% of articles. The unique set of terms is the corpus vocabulary. Each column of the DTM corresponds to an element of the vocabulary.
14. Convert an article’s word list into a vector of counts for each term in the vocabulary. This vector is the row of the DTM corresponding to the article.

IA.B. Topic Model Estimation

As mentioned above, in principle, one could estimate ϕ and θ via maximum likelihood according to (1). However, in most text applications this is computationally unrealistic. Instead, we use Bayesian methods and approach estimation with the collapsed Gibbs sampler proposed by Xiao and Stibor (2010).²

The intuition behind the estimation procedure can be understood with minimal technical overhead. First, note that model (1) can be equivalently represented as a “generative” model; that is, as a set of sampling rules. Taking model parameters ϕ_k and θ_t as given,³ we can conceptualize writing an

²We refer interested readers to Steyvers and Griffiths (2007) for an especially transparent description of LDA Gibbs samplers. Asuncion et al. (2009) review and compare alternative estimation approaches such as variational Bayes and maximum a posteriori estimation. They conclude that, with proper hyperparameter tuning, all estimators perform similarly in terms of model fit.

³These are drawn in a preliminary step from the Dirichlet priors, $\theta_t \sim \text{Dir}(\alpha)$ and $\phi_k \sim \text{Dir}(\beta)$.

article in the context of the model. Consider the t^{th} article with N_t total words. How does the model “write” the first term in the article? First, it draws a topic at random from the list of all topics, where the probabilities of drawing each topic are given by the $K \times 1$ vector θ_t . The topic assignment of that first term, call it $z_{t,1}$, is distributed as a unit multinomial (i.e., categorical) with parameter θ_t . Suppose we happened to draw topic k for the first term’s topic. To then draw the first term itself, we randomly choose from the vocabulary of all terms, where the probability of drawing any given term is governed by the $V \times 1$ vector ϕ_k , which is the topic-specific term probability distribution. Call this first term $x_{t,1}$, which is distributed as a unit multinomial with parameter ϕ_k . This process is repeated for each subsequent term in the article, a total of N_t times, at which point the model has authored the complete article and stops. In distribution notation, each term i in the article is given by

$$x_{t,i} \sim \text{Mult}(\phi_{z_{t,i}}, 1), \quad z_{t,i} \sim \text{Mult}(\theta_t, 1).$$

The Gibbs sampling estimation procedure searches for parameter values ϕ_k and θ_t that simulate articles most similar to those found in our news data set.⁴

The estimated attention that article t allocates to topic k can be repre-

⁴We take 300 iterations of the Gibbs sampler over the entire corpus, and use hyperparameters of $\beta = 1$ and $\alpha = 1$.

sented as the frequency with which its terms are assigned to k ,

$$\hat{\theta}_{t,k} = \frac{\sum_{i=1}^{N_t} \mathbb{I}(\hat{z}_{t,i} = k)}{\sum_{q=1}^K \sum_{i=1}^{N_t} \mathbb{I}(\hat{z}_{t,i} = q)}, \quad (\text{IA.1})$$

where $\hat{z}_{t,i}$ are topic assignments from the estimated model. Likewise, estimates of the topics themselves are determined by the number of times each term v in the vocabulary is assigned to topic k (aggregated over all articles):⁵

$$\hat{\phi}_{k,v} = \frac{\sum_{t=1}^T \sum_{i=1}^{N_t} \mathbb{I}(x_{t,i} = v) \mathbb{I}(\hat{z}_{t,i} = k)}{\sum_{m=1}^V \sum_{t=1}^T \sum_{i=1}^{N_t} \mathbb{I}(x_{t,i} = m) \mathbb{I}(\hat{z}_{t,i} = k)}. \quad (\text{IA.2})$$

Finally, while $\hat{\theta}_t$ describes the allocation of news attention to each topic in article t , we are often interested in how the *WSJ* as a whole allocates attention over time. We measure the overall news attention at a point in time τ (e.g., a given day or month) by summing over all articles at that time. Thus, *WSJ* attention allocation to topic k in month τ is

$$\hat{\theta}_{\tau,k} = \frac{\sum_{t \in \tau} \sum_{i=1}^{N_t} \mathbb{I}(\hat{z}_{t,i} = k)}{\sum_{t \in \tau} \sum_{q=1}^K \sum_{i=1}^{N_t} \mathbb{I}(\hat{z}_{t,i} = q)}. \quad (\text{IA.3})$$

Our model reportage focuses on estimates of the topics, $\hat{\phi}_k$, and estimates of attention allocation across topics, $\hat{\theta}_t$.

⁵As shown in (IA.1) and (IA.2), we choose to not incorporate priors in our θ and ϕ estimates. Griffiths and Steyvers (2004) show that priors can be easily incorporated as

$$\hat{\theta}_{t,k} = \frac{\sum_{i=1}^{N_t} \mathbb{I}(\hat{z}_{t,i} = k) + \alpha}{\sum_{q=1}^K \sum_{i=1}^{N_t} \mathbb{I}(\hat{z}_{t,i} = q) + K\alpha}, \quad \hat{\phi}_{k,v} = \frac{\sum_{t=1}^T \sum_{i=1}^{N_t} \mathbb{I}(x_{t,i} = v) \mathbb{I}(\hat{z}_{t,i} = k) + \beta}{\sum_{m=1}^V \sum_{t=1}^T \sum_{i=1}^{N_t} \mathbb{I}(x_{t,i} = m) \mathbb{I}(\hat{z}_{t,i} = k) + V\beta}.$$

IA.C. Topic Key Terms

Table IA.I. Topic Key Terms

Metatopic label	Topic label	Key terms
∞	Labor/Income	Executive pay <i>bonus, base salary, total compensation, pay package, compensation package, compensation committee, restrict stock, tyco, executive compensation, bonuses, stockoption, executive pay, stock option, compensation, annual salary, tyco international, exercise option, severance package, severance payment, exercise price</i>
		Job cuts <i>job cut, work force, fulltime employee, layoff, eliminate job, attrition, cut job, employment opportunity, employee work, unemployed, workforce, find job, parttime, lost job, temporary worker, retraining, job loss, skill worker, fulltime, humanresources</i>
		Unions <i>uaw, teamsters, relation board, union member, national labor, labor relation, picket line, worker union, union leader, nonunion, union official, international union, walkout, collective bargain, strikers, teamsters union, picket, union worker, labor agreement, contract talk</i>
		Health insurance <i>hmo, health plan, health coverage, health care, manage care, blue cross, health insurance, care act, blue shield, healthcare system, health maintenance, humana, affordable care, nurse home, medical cost, healthmaintenance, medicaid program, health insurer, maintenance organization, health system</i>
		Pensions <i>pension plan, employee retirement, benefit guaranty, pension, state pension, retirement system, retirement save, pension liability, retirement benefit, pension system, pension benefit, retirement, underfunded, retirement plan, retirement fund, retiree, public pension, retirement program, pension fund, retirement age</i>
		Government budgets <i>spend cut, deficit reduction, grammrudman, federal budget, balance budget, budget plan, cbo, house budget, entitlement program, budget, fiscal budget, federal spend, budget process, budget proposal, cut spend, budget office, domestic spend, congressional budget, budget cut, budget gap</i>
		Fees <i>fee paid, pay fee, fee, charge fee, fee charge, higher fee, receive payment, license fee, company pay, upfront, annual fee, reimburse, total cost, paid, extra cost, cash payment, company paid, additional cost, administrative cost, cover cost</i>
Management		Taxes <i>irs, tax rate, deduction, income tax, tax bill, irss, capitalgains, capitalgains tax, tax court, taxable income, corporate tax, tax law, tax system, marginal tax, tax expert, tax deduction, minimum tax, incometax rate, top tax, tax burden</i>
	Connecticut	<i>stamford conn, stamford, greenwich conn, conn, haven conn, clark, greenwich, norwalk, norwalk conn, westport, wallace, moore, kelley, westport conn, allen, murphy, evan, robinson, cunningham, johnson</i>
	C-suite	<i>executive officer, chief operate, operate officer, financial officer, chief financial, chairman president, officer john, officer robert, company chief, administrative officer, named chief, chief administrative, chairman chief, officer told, officer david, cfo, officer, remain chairman, act chief, finance chief</i>

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Table IA.I – *Continued from previous page*

Metatopic label	Topic label	Key terms
Trans/Defense/Local	Mid-level executives	<i>vice, vice president, executive vice, senior vice, named senior, named vice, named executive, president finance, president market, group vice, company vice, vice chairman, named president, group executive, newly create, business development, assistant vice, general manager, named, group president</i>
	Management changes	<i>succession plan, top job, departure, management shakeup, management style, shakeup, succession, heir apparent, top post, daytoday operation, nonexecutive chairman, successor, top management, management structure, daytoday, resignation, nonexecutive, management committee, management team, executive suite</i>
	Natural disasters	<i>katrina, quake, hurricane katrina, fishermen, coast guard, storm, emergency management, tsunami, earthquake, boat, hurricane, natural disaster, fisheries, baton rouge, salmon, beaches, mississippi river, debris, rita, dam</i>
	Police/crime	<i>inmates, police officer, violent crime, police department, gang, police, homicide, firearms, robbery, gunman, police chief, murder, traffickers, offenders, police arrest, death penalty, riot police, gun, heroin, sheriff</i>
	Mid-size cities	<i>kansas city, fla, lauderdale, palm beach, salt lake, lauderdale fla, lake city, fort lauderdale, beach fla, west palm, jacksonville, tampa, tampa fla, raton fla, raton, jacksonville fla, boca, fort, boca raton, tucson ariz</i>
	NY politics	<i>blasio, cuomo, gov andrew, mayor bill, mayor michael, city council, andrew cuomo, city mayor, mayor, city hall, mayoral, borough, councilman, york city, bronx, york gov, staten island, staten, city official, long island</i>
	Rail/trucking/shipping	<i>railroad, norfolk southern, conrail, csx, rail, union pacific, amtrak, burlington northern, norfolk, freight, ship company, truckers, shippers, truck company, transit, truck driver, port, railway, locomotive, uber</i>
	California	<i>los angeles, angeles, los, francisco, san francisco, diego, san diego, orange county, santa monica, angeles county, san, oakland, bay area, santa barbara, santa ana, los angelesbased, monica calif, angelesbased, northern california, san antonio</i>
	Rental properties	<i>square feet, square foot, vacancy rate, tenant, office space, landlord, office tower, squarefoot, retail space, highrise, apartment build, rent, redevelopment, tower, condo, midtown, square, time square, downtown, apartment</i>
	Disease	<i>disease control, aid virus, cdc, breast cancer, breast, infectious disease, heart disease, virus, flu, diagnosed, implants, immune system, infected, immune deficiency, deficiency syndrome, acquire immune, infectious, symptom, infection, world health</i>
	US defense	<i>rumfeld, missile defense, marine corps, joint chief, donald rumfeld, strategic defense, defense secretary, naval, senate arm, defense official, military service, cruise missile, defense initiative, nuclear missile, maj, pentagon official, arm service, military personnel, aircraft carrier, armor</i>

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Metatopic label	Topic label	Key terms
Government	Pharma	<i>pfizer, generic, bristolmyers, lilly, generic version, drug company, generic drug, eli lilly, novartis, drug maker, scheringplough, pharmaceutical industry, genentech, beecham plc, generic competition, roche, bristolmyers squibb, astrazeneca, squibb, drug price</i>
	Aerospace/defense	<i>force contract, lockheed, mcdonnell, mcdonnell douglas, northrop, pratt whitney, grumman, lockheed martin, airbus, general dynamic, european aeronautic, aeronautic defence, northrop grumman, martin marietta, defence space, raytheon, plane maker, airbus industrie, defense contractor, jet engine</i>
	Automotive	<i>auto maker, domestic car, import car, ford, sportutility, chevrolet, chrysler, mazda, toyota, eestimated, car maker, small car, sportutility vehicle, japanese auto, car market, honda, sedan, vehicle sale, minivan, ford motor</i>
	Airlines	<i>airline, texas air, twa, american airline, airways, pilot association, pilot union, flight attendant, load factor, northwest airline, usair, ual corps, unit airline, delta air, air line, line pilot, america west, delta, amr corps, continental airline</i>
	Retail	<i>walmart, department store, store open, walmart store, penney, kmart, retail analyst, comparablestore, bentonville, bentonville ark, largest retailer, neiman marcus, bloomingdales, specialty store, costco, discount store, macy, comparablestore sale, departmentstore, neiman</i>
	Political contributions	<i>donation, political contribution, philanthropy, campaign contribution, charity, nonprofit organization, nonprofit group, philanthropic, donor, donate, action committee, pac, nonprofit, political action, foundation, charitable, federal election, campaignfinance, advocacy, fundraising</i>
	Regulation	<i>propose rule, rule govern, rule require, final rule, federal regulation, consumer protection, regulation, rule change, current rule, protect consumer, federal rule, rulemaking, report requirement, regulate, government regulation, ftc, stricter, propose legislation, requirement, state regulation</i>
	Environment	<i>epa, emission, protection agency, environmental protection, carbon dioxide, dioxide, greenhouse gases, clean air, environmental, environmental group, greenhousegas, greenhousegas emission, air act, superfund, global warm, pollutants, hazardous waste, environmentalists, greenhouse, carbon</i>
	Private/public sector	<i>private sector, private enterprise, private business, public sector, private firm, private company, privatization, government aid, privatize, taxpayer money, private capital, privatesector, federal government, government money, government subsidy, government intervention, local government, private owned, state enterprise, government plan</i>
	State politics	<i>rhode island, state legislature, state include, west virginia, state legislator, schwarzenegger, north carolina, rhode, washington state, state lawmaker, legislature, state official, carolina, state house, republican gov, state level, virginia, outofstate, state capital, state budget</i>
	Watchdogs	<i>inspector general, gao, account office, general account, accountability office, government accountability, page report, report conclude, inspector, review process, house subcommittee, release report, correspondence, issue report, oversight, internal review, documentation, receive letter, wrote letter, examination</i>

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Metatopic label	Topic label	Key terms
Courts	Utilities	<i>nrc, megawatt, nuclear regulatory, pge, nuclear plant, electricity, electric utility, public utility, utility commission, power company, power plant, energy regulatory, power generation, grid, edison, utility hold, nuclear power, utility company, power light, nuclearpower plant</i>
	Safety administrations	<i>safety board, traffic safety, highway traffic, safety administration, transportation safety, osha, national highway, occupational safety, health administration, accident, safety standard, national transportation, air bag, safety, crashes, safety problem, safety issue, occupational, fatalities, firestone</i>
	National security	<i>cybersecurity, security agency, hacker, intelligence committee, fbi, cia, espionage, homeland security, security expert, senate intelligence, hacking, intelligence agency, spy, fbi agent, encryption, surveillance, security number, central intelligence, intelligence, breaches</i>
	Justice Department	<i>spitzer, antitrust division, general eliot, receive subpoena, eliot spitzer, department antitrust, department investigation, justice department, internal investigation, general office, investigate allegation, subpoena, cooperate fully, deputy attorney, probe, york attorney, attorney general, eliot, assistant attorney, criminal probe</i>
	Indictments	<i>indictment, plead guilty, plead, plea agreement, mail fraud, face maximum, guilty plea, wire fraud, prosecutor allege, guilty, federal grand, indicted, grand jury, conspiracy, felony count, allege scheme, defense attorney, criminal trial, acquitted, security fraud</i>
Political Leaders	Courts	<i>supreme court, appeal court, scalia, federal appeal, chief justice, high court, justice william, william rehnquist, antonin, court rule, court decision, justice john, supreme, lowercourt, antonin scalia, anthony kennedy, justice antonin, threejudge, oral argument, threejudge panel</i>
	Lawsuits	<i>punitive damage, suit file, classaction, plaintiff lawyer, class action, suit allege, plaintiff attorney, lawsuit file, compensatory damage, classaction suit, suit seek, file suit, classaction status, seek classaction, damage award, suit, unspecified damage, plaintiff, file lawsuit, seek unspecified</i>
	Clintons	<i>lewinsky, whitewater, rodham clinton, hillary rodham, rodham, clinton administration, president clinton, independent counsel, impeachment, janet reno, clinton, starr, clinton plan, house counsel, rubin, watergate, robert rubin, bill clinton, paula, reno</i>
	US Senate	<i>majority leader, senate bill, daschle, house bill, hastert, senate majority, appropriations committee, senate floor, senate version, gop leadership, lott, minority leader, senate appropriations, leader trent, senate democrat, house floor, committee chairman, baucus, leader harry, house gop</i>
	Reagan	<i>contra, reagan administration, president reagan, shultz, george shultz, state george, regan, irancontra, james baker, secretary james, nicholas brady, reagan, secretary nicholas, assistant secretary, nicaraguan, meese, baker, carter administration, deputy assistant, sandinista</i>
	Bush/Obama/Trump	<i>obama administration, trump administration, president donald, president barack, bush administration, president dick, president bush, president joe, house press, barack, barack obama, bush plan, president obama, obama, cheney, dick cheney, president george, senior administration, bush, presidentelect</i>

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Metatopic label	Topic label	Key terms
International Affairs	Elections	<i>sen obama, romney, sen mccain, dukakis, mitt romney, democratic primary, republican primary, mitt, gop primary, run mate, republican presidential, gop candidate, caucuses, republican nominee, senate race, republican candidate, democratic candidate, rubio, marco rubio, democratic nominee</i>
	European politics	<i>opposition party, socialist party, ldp, parliamentary election, centerleft, parliamentary, rule party, centerright, rule coalition, main opposition, opposition politician, liberal party, conservative party, opinion poll, national election, party member, upper house, coalition government, national assembly, referendum</i>
	Middle east	<i>palestinian, gaza, hamas, serb, gaza strip, israeli, plo, israel, hezbollah, palestinian authority, arafat, mubarak, assad, milosevic, west bank, palestinian leader, netanyahu, israeli official, damascus, lebanese</i>
	Nuclear/North Korea	<i>pyongyang, north korea, north korean, kim jong, nuclear program, daewoo, korean, seoul south, jong, kim, iran nuclear, nuclearweapons program, nonproliferation, south korean, international atomic, korea, nuclearweapons, korean government, enrichment, korean company</i>
	Terrorism	<i>taliban, karzai, afghan, qaeda, kabul, pakistani, islamabad, musharraf, hamid karzai, afghan government, kashmir, president hamid, islamic state, osama, laden, pakistan, pervez, extremist group, hamid, suicide bomber</i>
	Iraq	<i>iraqi, saddam, saddam hussein, kurds, baghdad, iraqi leader, iraqi government, erdogan, iraqi official, ankara, northern iraq, tayyip, abu dhabi, dhabi, recep, recep tayyip, bahrain, turkish, hussein, persian</i>
	Russia	<i>yeltsin, moscow, putin, russian, kremlin, gorbachev, kiev, ukraine, russian president, boris yeltsin, president vladimir, medvedev, vladimir putin, ruble, ukrainian, mikhail gorbachev, vladimir, sergei, mikhail, gazprom</i>
	Trade agreements	<i>gatt, wto, tariff, trade representative, antidumping, trade pact, protectionist, trade official, general agreement, freetrade agreement, trade talk, freetrade, trade policy, bilateral trade, trade barrier, unfair trade, trade dispute, protectionism, trade war, trade deal</i>
	Latin America	<i>argentina, chavez, latin, latin america, latin american, hugo chavez, cuban, colombia, president hugo, paulo brazil, castro, silva, argentine, puerto, puerto rico, fidel castro, caracas, havana, fidel, luiz</i>
	Japan	<i>japanese company, tokyo japan, japan largest, japan biggest, shinzo, minister shinzo, abe, japanese business, strong yen, japan ministry, japanese consumer, japanese bank, japan bank, japan economy, japanese firm, osaka, japanese market, japan financial, big japanese, major japanese</i>
	Canada/South Africa	<i>south africa, quebec, canadian government, toronto stock, british columbia, calgary, canadian cent, mandela, ottawa, calgary alberta, johannesburg, canada largest, africa, vancouver, apartheid, vancouver british, nova scotia, canadian, scotia, alberta</i>

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Metatopic label	Topic label	Key terms
Entertainment	China	<i>guan, beijing, xinhua, chinese official, chinese government, chinese company, chinese, zhu, beijing china, china sea, chinese authority, chinese leader, jiang, wen, xinhua news, zhang, chinese bank, deng, guangzhou, china central</i>
	Southeast Asia	<i>jakarta, thailand, southeast asia, rupiah, southeast asian, suharto, indonesian, baht, indonesia, bhd, jakarta indonesia, thai, malaysia, malaysian, hutchinson whampoa, hutchison, philippine, pte, asia pacific, singapore dollar</i>
	Germany	<i>east german, helmut kohl, bonn, chancellor helmut, helmut, german company, east germany, german official, west germany, chancellor gerhard, germany largest, gerhard, german chancellor, german bank, gmbh, berlin, supervisory board, merkel, german economy, german government</i>
	France/Italy	<i>generale, chirac, societe generale, french government, sarkozy, societe, french bank, belgium, francois, french company, cie, italian, french, belgian, jacques chirac, italian bank, president francois, french president, president jacques, italy</i>
	UK	<i>gbp, scotland, hold plc, scotland group, plc british, british bank, british company, scottish, group plc, unit kingdom, minister margaret, london british, lloyd bank, plc, westminster, bank plc, westminster bank, edinburgh, gordon brown, britons</i>
	Music industry	<i>album, record label, song, record company, music industry, universal music, music group, music company, emi, musicians, music, concert, warner music, jazz, rap, band, singer, musical, singing, promoter</i>
	Broadcasting	<i>clear channel, broadcaster, broadcast, cable channel, broadcast network, television station, cos nbc, nbc, viewership, liberty media, news network, television network, cable news, primetime, national broadcast, cbss, news channel, espn, channel, corps fox</i>
	Publishing	<i>barnes noble, circulation, time mirror, random house, magazine publisher, daily news, gannett, newspaper publisher, daily newspaper, chicago tribune, newsweek, publisher, publishes, barnes, journalism, newspaper, simon schuster, tribune, publish company, manage editor</i>
	Marketing	<i>saatchi, interpublic, omnicom, dbb, rubicam, saatchi saatchi, bbdo, wpp, young rubicam, publicis, omnicom group, wpp group, walter thompson, interpublic group, leo burnett, ogilvy mather, foote cone, mccannerrickson, ogilvy, mediabuying</i>
	Movie industry	<i>boxoffice, hollywood, paramount picture, motion picture, dreamworks, studio, sony picture, eisner, hollywood studio, mattel, theme park, movie, disney, hasbro, movie studio, columbia picture, animation, marvel, sequel, theatrical</i>
Social/Cultural	Economic ideology	<i>letter letter, relate letter, hoover institution, moral, liberalism, enterprise institute, american enterprise, morality, hoover, franklin roosevelt, capitalism, roosevelt, civilization, rightly, truman, freedom, myth, selfinterest, welfare state, courage</i>
	Schools	<i>public school, private school, high school, school, curriculum, test score, teacher union, school board, student, school district, school system, educators, classroom, teacher, elementary, elementary school, graduation, faculty, education department, graduate</i>

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Table IA.I – *Continued from previous page*

Metatopic label	Topic label	Key terms
Challenges	Sales call	<i>conference room, salesman, salesmen, boss, phone call, hotline, colleague, receive call, desk, stunned, huddled, caller, wanted, summoned, knew, began work, replied, salespeople, staffer, faceto face</i>
	Cultural life	<i>dog, cowboy, deer, jacket, shirt, sweat, tshirt, cat, wear, shower, horse, purple, dirt, dusty, wooden, baseball, bike, tennis, stadium, shiny</i>
	Arts	<i>sothebys, museum, auction house, art, collector, gallery, portrait, exhibit, thcentury, artistic, paint, statue, exhibition, monument, opera, auctioneer, collection, historian, archive, tribute</i>
	Immigration	<i>immigrant, illegal immigrant, migrant, census, census bureau, immigration law, undocumented, immigration, citizenship, illegal immigration, migration, legal status, deportation, poverty, inequality, population growth, residency, latino, hispanic, food stamp</i>
	Positive sentiment	<i>awful lot, pretty good, spend lot, good time, awful, good idea, making money, people feel, good deal, big money, feel good, pretty, good reason, lot, spend money, lose money, luck, feel comfortable, hard time, bad idea</i>
	Humor/language	<i>phrase, fool, word, humor, polite, translation, stupid, funny, emotion, tongue, honest, dumb, jargon, metaphor, honesty, silly, joke, truth, answer, fascinating</i>
	Gender issues	<i>female, young women, mother, daughter, divorce, father, young children, sexual harassment, family member, wives, male, husband, women, dad, child, married, young adult, mom, marry, sex</i>
Challenges	Changes	<i>major shift, shift, fundamental change, big change, major change, significant change, rethink, rapid change, direction, change, dramatic, emphasis, turn point, chang, reversal, urgency, move closer, move quick, reshape, focus</i>
	Key role	<i>key role, play role, play key, play major, major role, important role, big role, play big, lead role, active role, central role, role, bigger role, play, close relationship, potential conflict, play part, relationship, play field, influence</i>
	Problems	<i>problem face, biggest problem, big problem, matter worse, major problem, real problem, bigger problem, similar problem, solve, fundamental problem, problem, predicament, mess, plague, debacle, fiasco, disastrous, financial problem, problem include, inability</i>
	Challenges	<i>uphill battle, uphill, face tough, face pressure, battle, intense pressure, challenge face, status quo, quo, face challenge, increase pressure, tactic, fight, bruising, tough decision, backfire, backlash, bitter, face grow, persuade</i>
Challenges	Small possibility	<i>door open, time frame, frame, door, move quick, left open, wait, leave, ready, takes place, open, exit, left, stay, lock, quick, bringing, hand, final, rush</i>
	Spring/summer	<i>early march, late summer, close watch, late march, watch close, early sign, early stage, summer, watch, show sign, dash, early, spring, stage, warn sign, midmarch, turn point, slow start, began, fade</i>

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Metatopic label	Topic label	Key terms
Science/Language	Long/short term	<i>long term, short term, long run, longer term, long time, term limit, short run, long period, long haul, long, term, long history, long overdue, long list, longer, short time, short period, shorter, short, sooner</i>
	Research	<i>survey conduct, respondents, survey show, survey, survey release, methodology, data compile, statistical, gallup, study release, data show, percent, study show, compile, satisfaction, measurement, statistics show, study suggest, snapshot, sample</i>
	Scenario analysis	<i>financial analysis, commentary, worstcase scenario, worstcase, analysis, foreseeable, foreseeable future, conventional wisdom, imply, scenario, probability, adverse effect, plausible, theoretical, certainty, wisdom, consequence, assumption, negative effect, theory</i>
	Programs/initiatives	<i>program aim, goal, broader effort, top priority, pilot program, initiative, ambitious, achieve, program design, high priority, priority, revamp, aim, important step, remain committed, ambitious plan, objective, making progress, progress, development program</i>
	Biology/chemistry/physics	<i>gene, dna, mice, molecules, biology, scientist, physics, molecular, national laboratory, genetic, fusion, lab, massachusetts institute, physicist, mit, cell, research team, chemistry, nobel prize, experiment</i>
	Space program	<i>nasa, space administration, national aeronautic, space station, space shuttle, spacecraft, astronauts, orbit, booster, mann, shuttle, rocket, unmanned, launches, moon, satellite, robot, space, drone, aeronautic</i>
	Systems	<i>task force, task, current system, centralized, decentralized, system work, complexity, efficient, inefficiency, workable, cumbersome, innovation, antiquated, inefficient, function, rigid, flexible, solution, mechanism, find ways</i>
	Size	<i>big chunk, chunk, large chunk, smaller, bigger, smaller company, rough double, rely heavily, size, rapid grow, larger company, larger, bigger share, big reason, grow rapid, huge amount, larger share, grow number, grow fast, big part</i>
Activism/Language	Wide range	<i>wide range, vary wide, vast majority, broad range, wide variety, large number, limit number, large portion, vary, wide, small portion, unusual large, large amount, significant number, significant portion, small number, significant amount, wide considered, vast, range</i>
	Activists	<i>icahn, carl icahn, investor carl, getty, pennzoil, carl, texaco, white plain, source close, holmes, interfere, forestall, attempt, source, phillip, plain, nybased, control, intention, apparent</i>
	Announce plan	<i>announce plan, plan announce, unveil plan, announce earlier, previous announce, yesterdays announcement, announce restructure, company announce, outline plan, latest move, announce, plan include, restructure plan, move aim, current plan, announcement, company plan, plan call, expansion plan, unveil</i>

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Metatopic label	Topic label	Key terms
Negotiations	Major concerns	<i>raise concern, major concern, express concern, concern, increase concern, voice concern, concern raise, cit, held, addition, grow concern, significant, continue, remain, considered, represent significant, add, fully, represent major, lead</i>
	Futures/indices	<i>spot index, dollar yen, brother treasury, commodity oil, oil barrel, treasury index, future index, bond lehman, stock volume, jonesaig future, vol share, nasdaq vol, vol, jonesaig, bond shearson, market stock, share nasdaq, volume share, yld, industrial rose</i>
	Corrections/amplifications	<i>amplifications, correction amplifications, article correction, incorrect, news article, incorrect state, relate article, incorrect report, article, correction, misstated, york contribute, wsj, wsj sept, contribute, wsj march, source wsj, wsj research, money invest, washington contribute</i>
	Buffett	<i>buffett, berkshire, hathaway, berkshire hathaway, warren buffett, warren, omaha neb, omaha, neb, billionaire investor, brandon, cit, billionaire, significant, large part, continue, part due, charlie, held, substantial</i>
	Mexico	<i>mexican, salinas, mexican government, grupo, mexico, mexico city, ernesto, lopez, mexican peso, telefonos, president carlo, peso, enrique, nafta, raul, miguel, slim, gonzalez, luis, american free</i>
	Restraint	<i>temporary restrain, restrain order, postponement, postpone, delay, extension, indefinite, temporarily, expiration date, restrain, deadline, halt, cancel, temporary, extend, cancellation, schedule, resume, suspension, shutdown</i>
	News conference	<i>news conference, told news, press conference, told reporter, conference, press release, news release, remark, attendees, public appearance, reporter, conference call, public statement, statement release, issue statement, apology, transcript, industry conference, apologize, forum</i>
	Company spokesperson	<i>call seek, seek comment, return call, spokesman confirm, return phone, spokesman decline, spokesman add, return telephone, company spokesman, comment spokesman, official decline, company spokeswoman, immediately respond, decline comment, spokeswoman decline, company decline, elaborate, unavailable, comment, spokeswoman</i>
	People familiar	<i>people familiar, familiar, person familiar, person close, people close, people involve, person, matter, individual familiar, executive familiar, potential deal, situation, briefed, earlier report, official familiar, potential sale, knowledgeable, discussion, unclear, remain unclear</i>
	Agreement reached	<i>definitive agreement, sign letter, sign definitive, definitive, reach definitive, agreement call, lone star, preliminary agreement, agreement include, agreement reach, disclose term, agreement sign, sign agreement, tentative agree, agreement, financial term, final agreement, license agreement, reach agreement, lone</i>
	Committees	<i>panel member, advisory committee, advisory panel, recommendation, proposal include, proposal call, advisory group, final decision, panel, vote unanimous, advisory, commission report, advisory board, proposal, committee member, public hear, recommend, commission approve, unanimous, final report</i>

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Metatopic label	Topic label	Key terms
Industry	Negotiations	<i>stick point, negotiator, negotiate table, side agree, negotiation, impasse, reach deal, reach compromise, resolve dispute, unresolved, talk aim, stumble block, negotiate, haggling, talk, contentious issue, start talk, hold talk, deadlock, resolve</i>
	Soft drinks	<i>coke, bottler, pepsi, softdrink, cola, cocacola, soft drink, pepsico, north america, beverage company, schweppes, soda, north american, pepper, beverage, fountain, cadbury schweppes, american operation, volume growth, cadbury</i>
	Small business	<i>smallbusiness, small business, smallbusiness owner, business owner, national federation, independent business, small firm, small company, business administration, mediumsize business, large corporation, mediumsize, midsize company, big business, entrepreneurship, big corporation, entrepreneur, business, business community, large company</i>
	Cable	<i>warner cable, time warner, aol time, warner, parsons, warner communication, levin, adelphia, aol, warner incs, bronfman, buy time, cox, turner, hbo, comcast, cable system, america online, armstrong, media giant</i>
	Fast food	<i>franchisee, taco bell, burger king, fastfood, rite aid, starbucks, pizza hut, mcdonald, taco, fastfood chain, pizza, rite, restaurant, restaurant chain, burger, wendy, fastfood restaurant, hut, sandwiches, hamburger</i>
	Competition	<i>goodyear, stiff competition, tire rubber, increase competition, fierce competition, goodyear tire, competitive advantage, competitive market, increase competitive, high competitive, intense competition, tire maker, competitive environment, competition, face stiff, competitive pressure, akron ohio, tough competition, akron, grow competition</i>
	Chemicals/paper	<i>union carbide, carbide, specialty chemical, georgiapacific, chemical company, pont, weyerhaeuser, chemical concern, chemical business, dupont, chemical maker, paper product, hercules, international paper, paper company, coatings, danbury, chemical industry, wilmington, forestproducts</i>
	Venture capital	<i>venture, joint venture, venture capitalist, venture capital, venturecapital firm, venturecapital, form joint, venture partner, venture fund, venturecapital fund, jointventure partner, joint, capitalist, startup company, jointventure, byers, startup, perkins, joint owned, form partnership</i>
	Tobacco	<i>tobacco, smokers, tobacco company, cigarette, philip morris, nicotine, tobacco industry, morris, cigarette maker, reynolds tobacco, philip morriss, altria, morriss, antismoking, marlboro, morris cos, bradstreet, dun, dun bradstreet, nabisco hold</i>
	Subsidiaries	<i>owned subsidiary, wholly owned, spinoff, itt, parent company, unit include, separate company, wholly, noncore, business unit, noncore asset, operate unit, spin, operation include, service unit, service division, spun, divestiture, subsidiary, core business</i>
	Credit cards	<i>mastercard, credit card, debit card, debit, creditcard company, card, creditcard issuer, american express, creditcard, express, creditcard business, check account, visa, teller machine, teller, payment system, hallmark, prepaid, express, merchant</i>

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Metatopic label	Topic label	Key terms
	Couriers	<i>postal service, fedex, postal, unit parcel, ups, parcel service, post office, federal express, parcel, mail, envelope, firstclass, memphis tenn, direct mail, memphis, downs, delivery system, airborne, mailorder, stamp</i>
	Foods/consumer goods	<i>sara lee, general mill, nestle, hershey, avon, campbell soup, diapers, detergent, cereal, toothpaste, heinz, quaker oats, ice cream, sara, information resource, unilever, quaker, yogurt, shampoo, chocolate</i>
	Insurance	<i>policyholders, life insurance, reinsurance, insurance department, insurance regulator, allstate, insurance commissioner, mutual life, life insurer, insurance business, insurance broker, propertycasualty, metlife, insurance unit, metropolitan life, insurance operation, life casualty, aig, lifeinsurance, aetna life</i>
	Luxury/beverages	<i>lvmh, vuitton, hennessy louis, louis vuitton, moet, moet hennessy, lvmh moet, beer, brewer, wine, coors, brewery, anheuserbusch cos, liquor, brew, anheuserbusch, scotch, whiskey, luxurygoods, budweiser</i>
	Casinos	<i>casino, atlantic city, marriott, hilton, carnival, hilton hotel, hotel chain, hotel room, inn, caesars, resort, mirage, hotel, hyatt, poker, casino operator, vegas, circus, hospitality, lodge</i>
	Revenue growth	<i>growth strategy, growth opportunity, annual revenue, boost revenue, growth area, company revenue, total revenue, company growth, revenue growth, growth potential, core business, generate revenue, sanford bernstein, post revenue, future growth, growth plan, company core, fastergrowing, revenue stream, increase revenue</i>
Technology	Internet	<i>facebook, web site, search engine, alibaba, yahoo, ebay, youtube, home page, internet user, online advertise, portal, socialnetworking, social network, internet company, web, twitter, web page, online, internet site, amazon</i>
	Mobile devices	<i>iphone, smartphone, android, handset, ipad, blackberry, mobile device, nokia, apple incs, mobile phone, tablet, rim, cellphone, qualcomm, ipod, steve job, mobile, galaxy, telefon, text message</i>
	Electronics	<i>playstation, nintendo, console, kodak, digital camera, lcd, flatpanel, xbox, cdrom, electronic art, videogame, photographic, video game, lenses, highdefinition, panasonic, compact disk, camera, photography, stereo</i>
	Phone companies	<i>att, sprint, longdistance, local phone, bell atlantic, phone company, baby bell, bell company, mci, longdistance company, regional bell, mccaw, phone service, qwest, longdistance service, bellsouth, nynex, telephone company, ameritech, nextel</i>
	Computers	<i>ibm, compaq, mainframe, dell, mainframe computer, machine corps, armonk, computer giant, packard, computer business, workstation, big blue, supercomputer, cray, compaq computer, computer company, computer associate, emc, computer market, business machine</i>

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Metatopic label	Topic label	Key terms
Oil & Mining	Software	<i>linux, window operate, ballmer, microsoft window, redmond wash, redmond, microsoft, oracle, software giant, lotus, software industry, ellison, bill gate, spreadsheet, netscape, novell, sap, adobe, window, java</i>
	Microchips	<i>amd, advance micro, memory chip, santa clara, semiconductor industry, clara, intel chip, intel, clara calif, chip maker, chip, memorychip, computer chip, micro device, micron, pentium, cisco, micro, national semiconductor, seagate</i>
	Steel	<i>steel price, steel industry, steelmaker, alcan, steel product, nucor, steel producer, steel, steel mill, steel company, aluminum, bethlehem steel, bethlehem, steelmaking, alcoa, usx, newsprint, smelter, raw material, metric</i>
	Mining	<i>bhp, rio tinto, tinto, newmont, bhp billiton, mining company, miner, mining, billiton, newmont mining, mining concern, sydney australia, anglo american, gold producer, goldmining, coal miner, mines, melbourne, anglo, australian</i>
	Machinery	<i>machine tool, manufacture plant, whirlpool, manufacture facility, construction equipment, manufacture operation, manufacture capacity, appliance maker, production facility, cummins, production line, factory, manufacture company, supply chain, deere, caterpillar, machinery, manufacture, wash machine, assembly line</i>
	Oil drilling	<i>exxon, drill, cubic feet, mobil, exxon mobil, drill rig, offshore oil, chevron, amoco, royal dutchshell, occidental, oil spill, dutchshell, cubic, arco, dutchshell group, deepwater, offshore drill, liquefied natural, conocophillips</i>
	Agriculture	<i>genetically modified, farm, growers, poultry, madcow, farmer, ranchers, tyson food, agriculture secretary, tyson, farmland, cow, usda, agriculture, agribusiness, cargill, tomatoes, agricultural, beef, fertilizer</i>
	Oil market	<i>opec, nonopec, opec member, oil minister, oil demand, world oil, brent crude, oil market, oil exporter, export country, petroleum export, refiner, oil supply, international energy, crude price, oil price, higher oil, high oil, oil export, oil refiner</i>
Economic Growth	Record high	<i>record level, highest level, reach record, hit record, previous record, highest, pentup demand, high level, pentup, alltime, alltime high, strong demand, record high, set record, higher level, helped fuel, remain strong, helped drive, helped push, surpass</i>
	Economic growth	<i>gdp growth, productivity growth, gdp, domestic product, gross domestic, economy grew, national product, economy expand, gross national, growth forecast, economic growth, business cycle, economic output, gnp, economic performance, growth rate, annual growth, slow economic, economy grow, economic expansion</i>
	Federal Reserve	<i>greenspan, alan greenspan, yellen, federal funds rate, federal funds, fomc, chairman alan, raise rate, janet yellen, inflation expectation, reserve chairman, quarterpoint, policy meet, market committee, target rate, federal open, bernanke, cut rate, rate hike, raise interest</i>

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Metatopic label	Topic label	Key terms
Corporate Earnings	European sovereign debt	<i>ecb, imf, eurozone, euro zone, monetary fund, international monetary, draghi, eurozone country, trichet, ecb president, jeanclaude trichet, greece, eurozone economy, greek, athens, austerity measure, bailout fund, european central, debt crisis, sovereigndebt</i>
	Recession	<i>economic downturn, steep drop, steep decline, downturn, bright spot, hardest hit, hit hard, weak economy, hit bottom, hard hit, economic slowdown, weak demand, sharp drop, economic slump, slump, weaken economy, steep, lowest level, gloomy, sharp decline</i>
	Product prices	<i>raise price, increase price, price reduction, price increase, price cut, wholesale price, higher cost, cut price, retail price, reduce price, price power, rise cost, higher price, cost increase, slash price, list price, discount price, price war, increase cost, price tag</i>
	Optimism	<i>cautious optimistic, remain optimistic, remain cautious, remain uncertain, uncertainty surrounding, positive sign, optimistic, overly optimistic, cautious, remain confident, remain skeptical, express confidence, positive impact, positive, optimism, uncertain, remain concern, uncertainty, remain wary, economic uncertainty</i>
	Macroeconomic data	<i>seasonal variation, durable good, construction spend, seasonal adjust, nonresidential, business inventory, nondurable, durable, conference board, jobless claim, manager index, factory order, purchase manager, supply management, manufacture activity, seasonal, initial claim, adjust annual, lead indicator, adjust basis</i>
	Profits	<i>net profit, profit rose, profit fell, firsthalf net, profit increase, profit jump, operate profit, profit climb, profit surge, pretax profit, profit dropped, profit rise, firsthalf, profit decline, group sale, group net, firsthalf profit, group profit, fullyear net, post rise</i>
	Revised estimate	<i>previous estimate, original estimate, preliminary estimate, revision, earlier estimate, revise downward, downward revision, initial estimate, revise, substantial increase, significant increase, estimate, official estimate, significant impact, substantial reduce, previous thought, government estimate, figure, figure show, revise wsj</i>
	Earnings losses	<i>loss include, net loss, discontinue operation, operate loss, loss widen, pretax charge, discontinue, aftertax charge, post loss, restructure charge, writedown, include loss, loss narrow, restructure cost, pretax loss, expect loss, loss total, report loss, impairment, special charge</i>
	Small changes	<i>typical require, average yield, decline slight, decrease, slight increase, remain unchanged, remain flat, sharp increase, part reflect, slight decline, increase reflect, continue decline, fell slight, slight, jumbo, slight higher, period, total number, rose slight, attribute</i>
	Financial reports	<i>ended sept, forma, pro forma, ended march, latest fiscal, march compare, current fiscal, full fiscal, ended, company fiscal, period ended, fiscal, sept, fiscalyear, pro, fiscal earn, report revenue, report sale, company expect, march</i>

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Metatopic label	Topic label	Key terms
Financial Markets	Earnings forecasts	<i>adjust earn, analyst earn, analyst poll, forecast earn, earn forecast, earningspershare, earn expectation, beat expectation, earn guidance, earn estimate, earn projection, earn growth, earn release, company forecast, analyst forecast, analyst expect, report earn, project earn, earn season, revenue forecast</i>
	Earnings	<i>cent dilute, dilute share, share revenue, share relate, callthomson financial, cent share, callthomson, call consensus, pershare figure, thomson financialfirst, share exclude, financialfirst call, financialfirst, earlier exclude, period revenue, income total, net income, dilute, share compare, exclude special</i>
	IPOs	<i>ipo, ipo market, initial public, public offer, ipo price, firstday, public stock, stock offer, roadshow, offer price, followon, price range, secondary offer, share offer, public market, initial offer, lockup, prospectus, company initial, lead underwriter</i>
	Bond yields	<i>yield rise, highyield bond, sell bond, yield fall, corporatebond, bond issue, treasury yield, issue bond, municipalbond, bond sale, higher yield, treasury debt, highyield, high yield, bond portfolio, bond deal, risk premium, muni, buy bond, governmentbond</i>
	Short sales	<i>short seller, short interest, short sell, borrow stock, borrow share, shortselling, shortsellers, short sale, sell borrow, seller, short position, prospective buyer, find buyer, potential buyer, buyer, short, investor sell, share sold, purchaser, foreign buyer</i>
	Small caps	<i>york smallcapitalization, poor smallcap, smallcapitalization stock, russell index, nyse rose, smallcap index, smallcap market, nyse fell, nasdaq gain, small-capitalization, nasdaq fell, nasdaq rose, nasdaq advance, smallcap stock, initiate coverage, nasdaq decline, smallcap, add cent, market perform, lost cent</i>
	Treasury bonds	<i>accept noncompetly, cusip, bid widen, price accept, noncompetitive tender, coupon equivalent, application accept, price rate, cusip number, bill mature, noncompetly, bid narrow, accept noncompetitive, rate coupon, bid accept, fourweek bill, minimum denominations, marketclearing, competitive tender, equivalent yield</i>
	International exchanges	<i>perspective geneva, dax stock, australia world, switzerland australia, germany hong, firstsection, firstsection volume, international perspective, kong switzerland, index calculate, world index, index eafe, mscip, stanley capital, zetra dax, timesstock exchange, firstsection issue, msci indexes, share index, japan france</i>
	Exchanges/composites	<i>composite trade, exchange composite, exchange trade, market composite, york stock, stock exchange, adrs, share close, board trade, composite, close unchanged, depositary, share rose, depositary receipt, board composite, trade share, american depositary, share fell, depositary share, stock close</i>
	Currencies/metals	<i>dollar british, frankfurt gold, home mkt, dollar french, london toronto, comex base, tokyo zurich, comex london, yen swiss, ounce comex, toronto paris, yen sterling, pound canadian, yen late, franc japanese, franc west, mark base, dollar stood, mark late, paris tokyo</i>
	Share payouts	<i>nyse symbol, share buyback, stock buyback, dividend increase, buyback, buyback program, share repurchase, share common, pay dividend, time estimate, stock repurchase, dividend yield, daily trade, dividend payment, dividend, special dividend, cash dividend, annual dividend, average daily, share share</i>

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Metatopic label	Topic label	Key terms
Buyouts & Bankruptcy	Bear/bull market	<i>bear market, bull market, stock pick, stockpicking, priceearnings ratio, growth stock, pricetoearnings ratio, bull, priceearnings, pricetoearnings, contrarian, stock portfolio, wilshire, foreign stock, ned, stockmarket, overvalued, investment strategist, big stock, beatendown</i>
	Commodities	<i>cocoa exchange, comex division, sugar cocoa, coffee sugar, soybean future, coffee future, lifeofcontract, pork belly, wheat future, york cotton, copper future, march contract, commodity market, william oneill, commodity fund, corn future, contract rose, contract fell, cotton exchange, future price</i>
	Trading activity	<i>action stock, fell bond, market action, otc trade, volume total, investor report, issue advance, board volume, index exclude, transportation average, otc market, otc, share chang, stock advance, board issue, bond gain, professional investor, decline stock, euro rose, active issue</i>
	Options/VIX	<i>option exchange, volatility index, board option, contract compare, put trade, option strategist, option market, viz, cboe, option trader, call trade, cftc, option trade, euronext, call option, trade contract, put option, option contract, future exchange, exchange official</i>
	M&A	<i>combine company, merge company, propose merger, combine entity, merger plan, merger talk, planned merger, merger, planned acquisition, antitrust concern, merge, deal close, propose acquisition, synergy, deal announce, deal valued, merger partner, antitrust regulator, acquire close, consolidation</i>
	Control stakes	<i>control stake, minority stake, majority stake, buy stake, abn, sell stake, remain stake, bought stake, minority shareholder, owns stake, acquire stake, hold stake, abn amro, amro, ownership stake, stake, company controlled, large stake, control interest, equity stake</i>
	Drexel	<i>drexel, leverage buyout, kohlberg kravis, drexel burnham, transaction finance, kohlberg, kravis robert, kravis, boesky, pickens, burnham lambert, burnham, buyout group, lambert, investor acquire, fund generate, ivan boesky, finance large, boone pickens, hicks</i>
	SEC	<i>sec file, sec, sec allege, sec official, commission file, sec enforcement, sec chairman, exchange commission, sec staff, sec investigation, sec spokesman, sec charge, arthur levitt, nonpublic information, chairman arthur, nonpublic, file show, violate security, enforcement division, sec rule</i>
	Bankruptcy	<i>bankruptcy, unsecured creditor, bankruptcy court, federal bankruptcy, bankruptcy code, chapter company, creditor lawsuit, bankruptcy judge, reorganization plan, bankruptcy protection, bankruptcy law, chapter bankruptcy, chapter, creditor committee, court protection, bankruptcy file, bankruptcy proceed, bankruptcy law protection, bankruptcy law, chapter protection</i>
	Corporate governance	<i>annual meet, shareholder meet, proxy contest, corporate governance, independent director, dissident shareholder, board seat, proxy battle, annual shareholder, proxy fight, shareholder vote, member board, shareholder group, institutional shareholder, board vote, bylaws, proxy material, governance, board meet, board member</i>
	Takeovers	<i>poison pill, hostile takeover, takeover bid, hostile bid, ashare offer, hostile tender, ashare tender, higher offer, takeover battle, takeover attempt, takeover offer, ashare bid, takeover defense, poisonpill, unsolicited bid, tender offer, unsolicited offer, takeover proposal, bid war, ashare</i>

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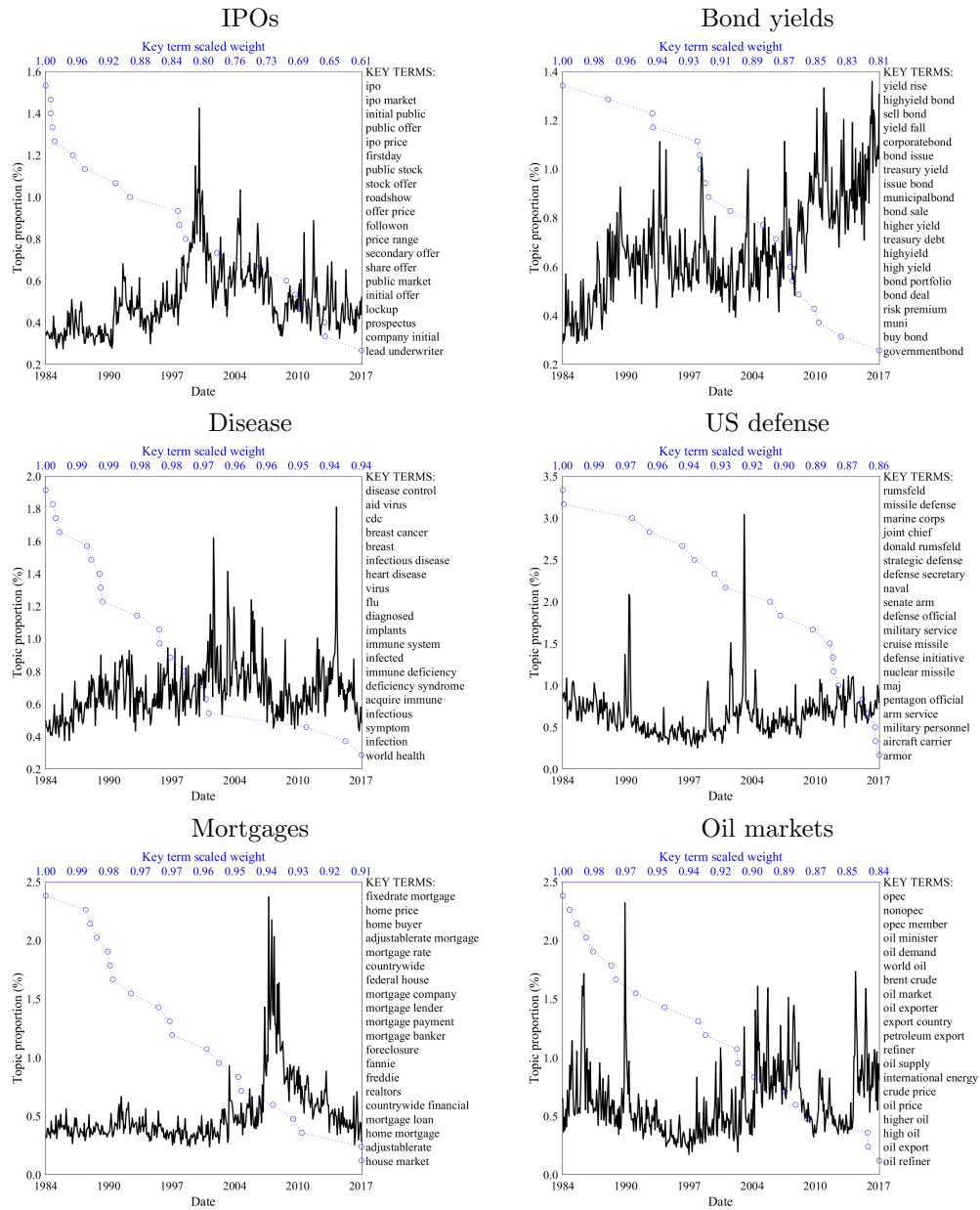
Metatopic label	Topic label	Key terms
Asset Managers/I-Banks	Real estate	<i>real estate, reit, estate investment, realestate investment, estate market, estate, estate developer, estate development, commercial real, realty trust, realty, investment trust, limit partnership, realestate company, commercial property, realestate, commercial realestate, realestate market, real, property</i>
	Convertible/preferred	<i>common share, preferred share, convertible preferred, common stock, class common, convertible debenture, company common, national overthecounter, convertible subordinate, preferred, preferred stock, overthecounter trade, class stock, stock outstanding, principal amount, corporate purpose, redeemable, share outstanding, convertible, newly issue</i>
	Mutual funds	<i>compound yield, simple yield, money fund, taxfree fund, assume reinvestment, taxable fund, stock fund, taxable money, fund company, taxexempt fund, moneymarket fund, closedend, company institute, noload, morningstar, fund rose, mutual fund, vanguard, fund industry, fund fell</i>
	Accounting	<i>andersen, account firm, kpmg, arthur andersen, fasb, deloitte, ernst young, audit firm, ernst, skadden, peat marwick, deloitte touche, skadden arps, meagher flom, arps, standard board, marwick, arps slate, slate meagher, meagher</i>
	Investment banking	<i>csfb, sachs group, goldman, suisse group, sachs, goldman sachs, credit suisse, suisse, ubs, morgan, dimon, group credit, stanley, morgan stanley, morgan chase, ubs warburg, swiss bank, investmentbanking, stanley dean, mack</i>
	Acquired investment banks	<i>salomon smith, smith barney, barney, merrill, bear stearns, stearns, merrill lynch, lynch, stearns cos, painewebber group, salomon, lufkin, donaldson lufkin, jenrette, donaldson, shearson, lufkin jenrette, brokerage unit, brokerage business, kidder</i>
	Private equity/hedge funds	<i>hedge fund, privateequity, privateequity firm, hedgefund, blackstone, carlyle, privateequity fund, hedgefund firm, hedgefund manager, blackstone group, tpg, management llc, bain capital, carlyle group, private equity, sovereignwealth fund, sovereignwealth, capital llc, soros, partner llc</i>
Banks	NASD	<i>nasd, customer account, security transaction, member firm, barred, disciplinary action, admit, disciplinary, misappropriated, deny, fine, security dealer, restitution, misrepresentations, authorization, consent, suspend, quotation, supervisory, unauthorized</i>
	Savings & loans	<i>fdic, thrift, loan association, federal deposit, federal save, deposit insurance, bank board, save loan, thrift industry, loan insurance, thrift supervision, rtc, save association, insurance fund, fail bank, federal insure, bank failure, save bank, loan bank, depositors</i>
	Nonperforming loans	<i>wells fargo, fargo, loanloss provision, bruyette, bankamerica, keefe bruyette, bruyette wood, nonperforming asset, loanloss, noninterest, interest margin, nonperforming, nonperforming loan, nationsbank, interstate bancorp, loanloss reserve, suntrust, pnc, loan growth, security pacific</i>
	Credit ratings	<i>debt rate, fitch rate, rate firm, rate service, creditwatch, rate agency, creditrating, taxes depreciation, junk status, notches, creditrating agency, senior debt, interest taxes, rate downgrade, operate cash, ebitda, negative implication, poor rate, rate company, fitch</i>

Continued on next page

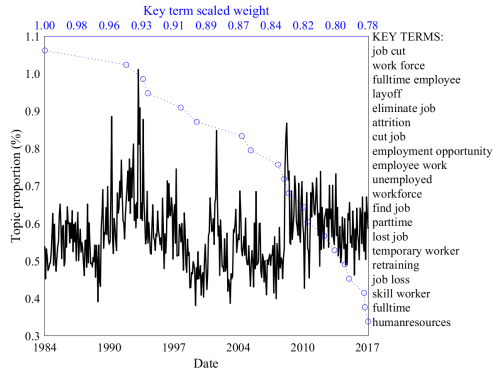
Table IA.I – Continued from previous page

Metatopic label	Topic label	Key terms
	Financial crisis	<i>stress test, relief program, large financial, trouble asset, financial system, creditdefault swap, financial regulator, creditdefault, financial firm, financial crisis, credit crisis, secretary henry, financial regulation, timothy geithner, geithner, government bailout, henry paulson, financial instrument, bank balance, basel</i>
	Bank loans	<i>syndicate loan, revolving credit, credit line, libor, credit agreement, loan agreement, revolving, bank loan, creditor bank, rescheduling, credit facility, principal payment, shortterm loan, repayment, loan program, corporate loan, bank debt, interbank offer, offer rate, london interbank</i>
	Mortgages	<i>fixedrate mortgage, home price, home buyer, adjustablerate mortgage, mortgage rate, countrywide, federal house, mortgage company, mortgage lender, mortgage payment, mortgage banker, foreclosure, fannie, freddie, realtors, countrywide financial, mortgage loan, home mortgage, adjustablerate, house market</i>

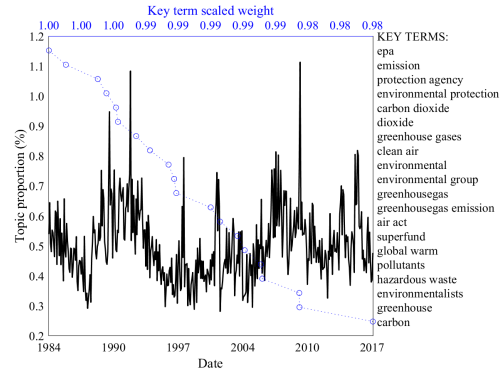
Figure IA.1. Topic Attention Proportions



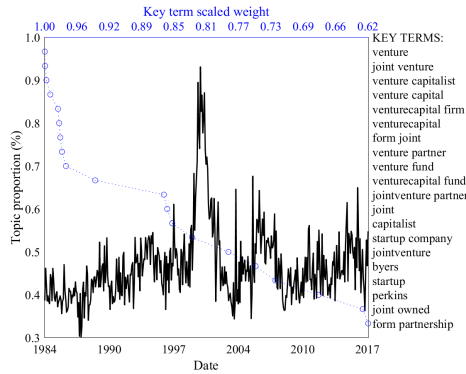
Job cuts



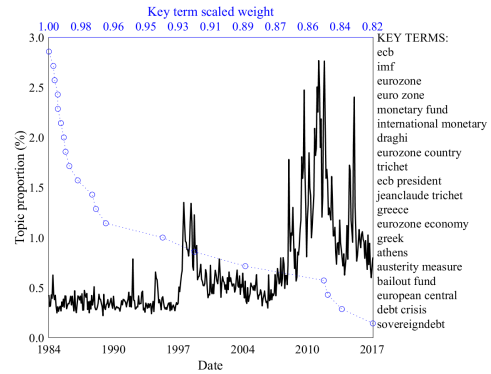
Environment



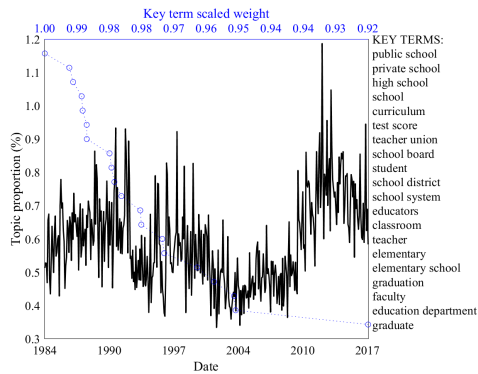
Venture capital



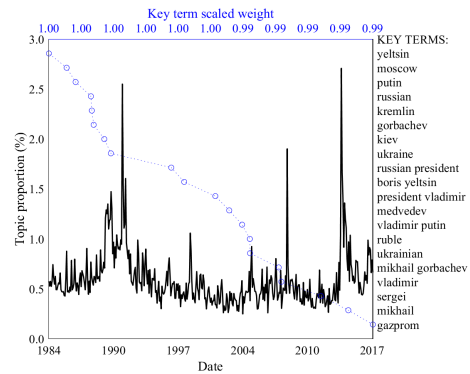
European sovereign debt

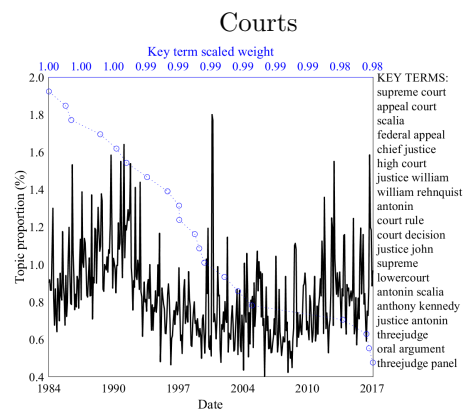
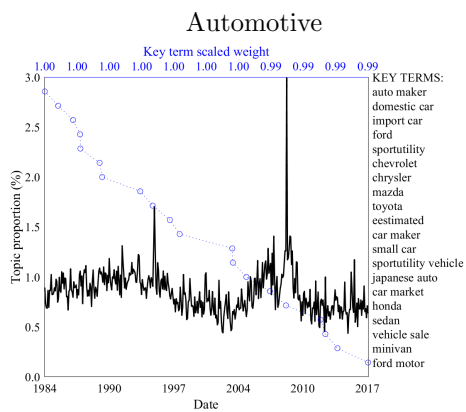
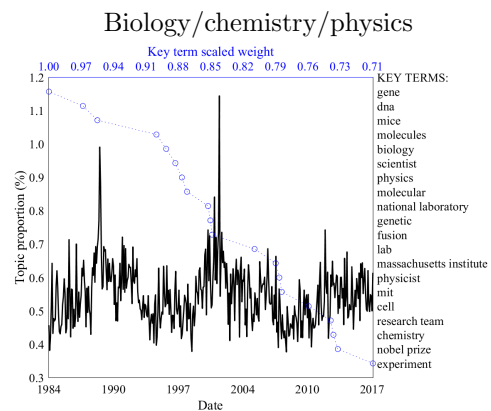
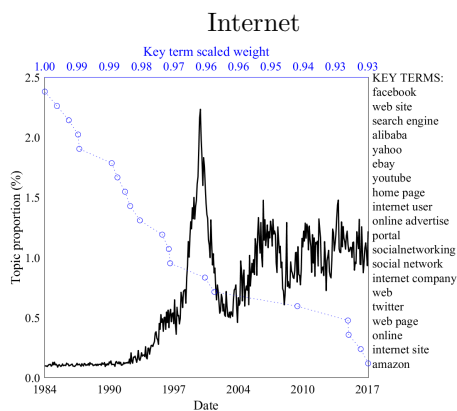
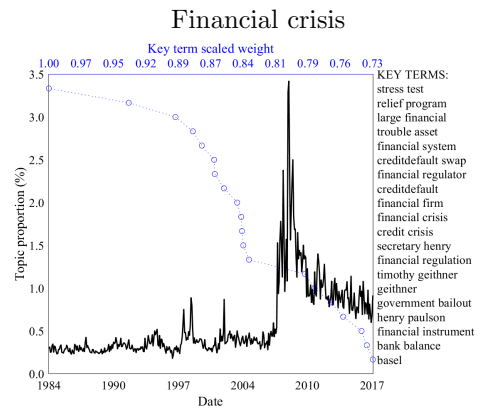
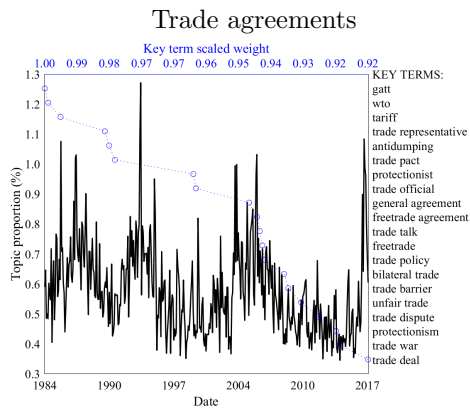


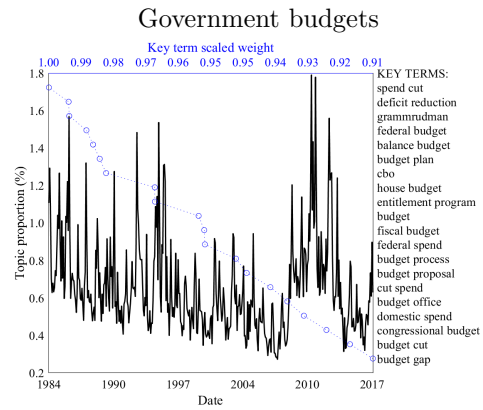
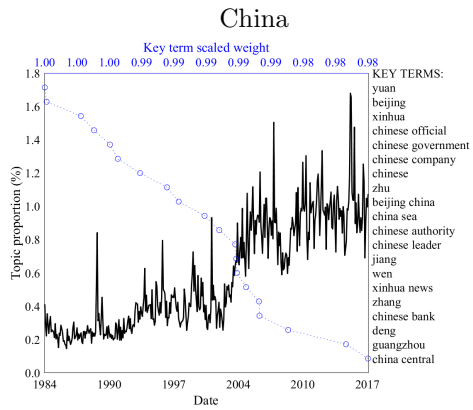
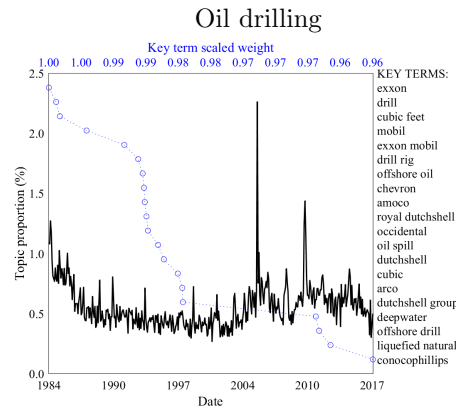
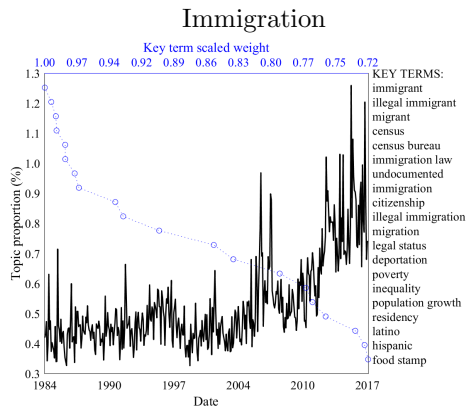
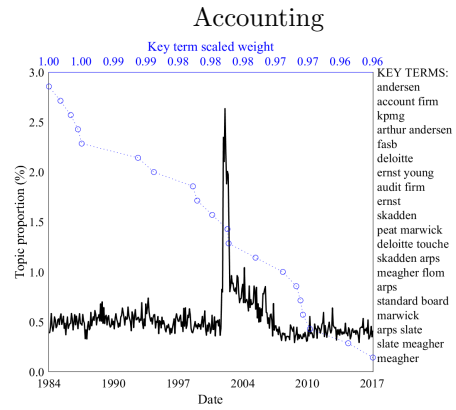
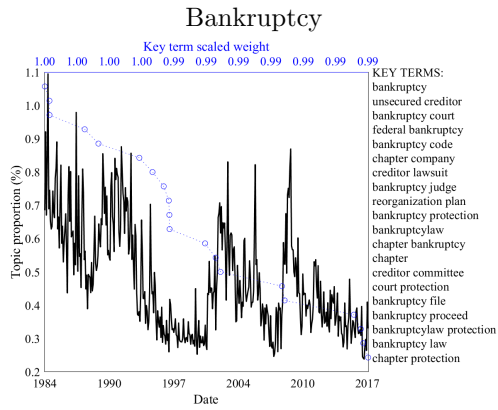
Schools



Russia



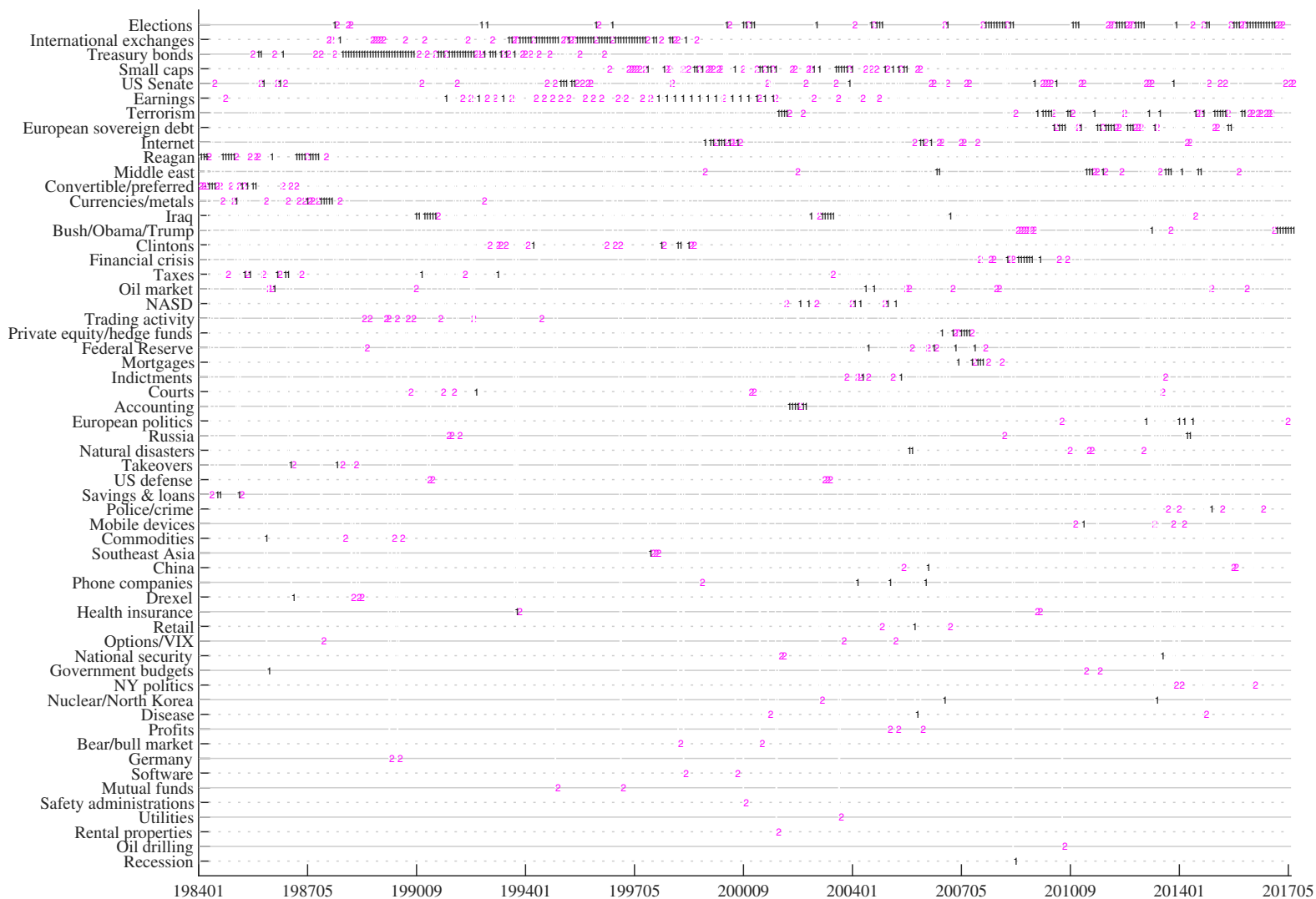




IA.D. Dominant Topics Over Time

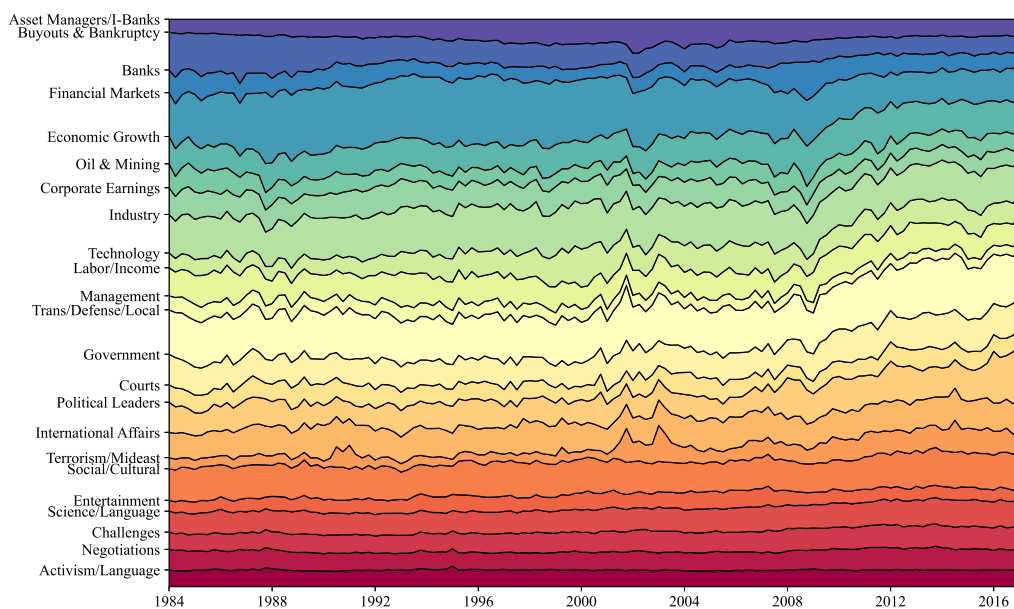
With 180 topics, it is difficult to isolate the focus of attention at each point in time in a single line plot overlaying many time series. To overcome this, Figure [IA.2](#) shows the top two topics receiving the most news attention each month. Dominant topics vary depending on exogenous factors (such as election cycles or natural disasters) and endogenous states of the economy (such as recessions and sovereign debt crises), with primary attention rotating between political, macroeconomic, industrial, and financial topics.

Figure IA.2. Dominant Topics By Month



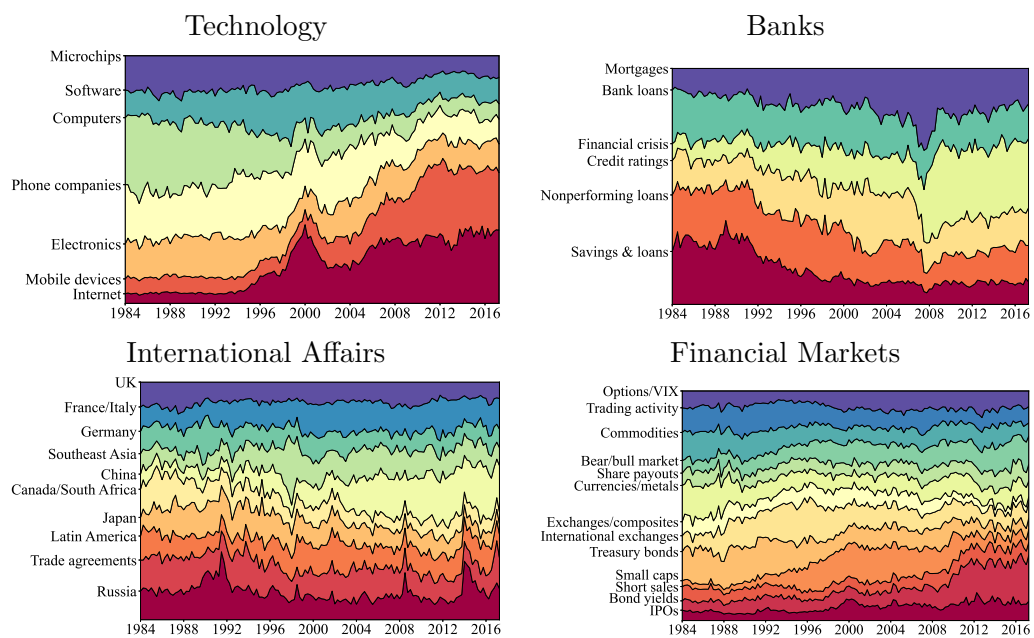
The figure reports the topics ranked 1 and 2 each month in terms of highest *WSJ* news attention proportion.

Figure IA.3. News Attention Across Metatopics



Stacked area plot of attention allocation to 23 metatopics based on the first-level clustering of baseline topics.

Figure IA.4. News Attention Within Metatopics



Stacked area plot of attention allocation to baseline topics within four of the first-level metatopics.

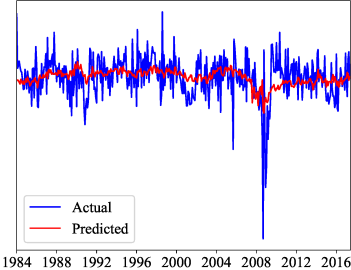
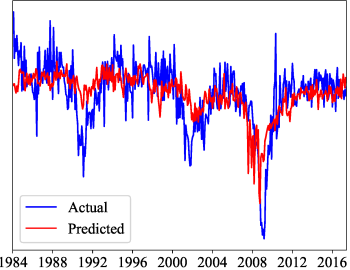
IA.E. Metatopics

Figure [IA.3](#) zooms out to show how attention behaves at the 23-metatopic level (the first level of aggregation on the right side of Figure [2](#)). The fraction of news attention allocated to metatopics is fairly stable over time. Yet there is a high degree of attention churn within metatopic clusters. For example, while attention to “technology” is stable over time in Figure [IA.3](#), its contents transition rather dramatically over time (upper left panel of Figure [IA.4](#)). It is dominated by mainframe and desktop computers at the start of the sample, is displaced by attention to software mid-sample, which is in turn displaced by mobile devices and the internet in the later part of the sample. Likewise, while attention to the “banks” metatopic is stable over time, the focus shifts from savings and loans in the 1980s to mortgages in the mid 2000s, and remains dominated by attention to financial crises even through the recent economic expansion. The “international affairs” metatopic includes steady attention to western Europe, but also shows episodic spikes in attention to Russia, a brief focus on Southeast Asia during its 1998 crisis, and the gradual emergence of China as a focal point by the end of the sample. The “financial markets” metatopic shifts focus from Treasuries and international markets in the first half of the sample, to small caps during the dotcom era and leading up to the financial crisis, after which bond yields become a focal point.

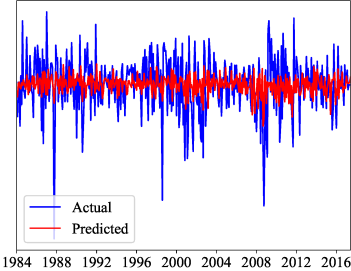
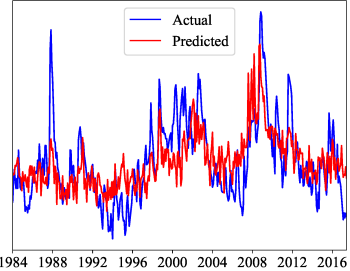
Metatopics provide additional dimension reduction, beyond that provided by LDA. It is therefore possible that metatopics are better able to explain variation in macroeconomic fundamentals or stock market prices. To investigate this possibility, in [Table IA.II](#) we repeat the analysis of [Table I](#) but using

Table IA.II. Reconstructing Macroeconomic Time Series Using Metatopics

Industrial Production Growth			Employment Growth		
Topic	Coeff.	p -val.	Topic	Coeff.	p -val.
International/Commodities	-0.23	0.04	International/Commodities	-0.39	0.00
Culture	0.14	0.55	Technology	0.30	0.20
Law/Courts	-0.11	0.95	Banks	-0.25	0.00
International Affairs	-0.09	0.02	Entrepreneurship	0.11	0.50
Entrepreneurship	0.07	0.77	National Events	0.08	0.93
R^2	0.12		R^2	0.37	

Market Returns			Market Volatility		
Topic	Coeff.	p -val.	Topic	Coeff.	p -val.
Economic Growth	-0.20	0.00	Banks	0.41	0.00
International Affairs	-0.20	0.14	Oil/Industry	-0.30	0.00
International/Commodities	-0.10	0.22	Entrepreneurship	-0.28	0.00
Law/Courts	0.10	0.35	International/Commodities	0.25	0.00
Culture	0.09	0.85	Economic Growth	0.08	0.06
R^2	0.11		R^2	0.44	

Five-regressor active-set regression estimates based on lasso selection with p -values adjusted for post-selection inference. Here, instead of topic attentions, we use the 23 higher-level metatopics as explanatory variables.

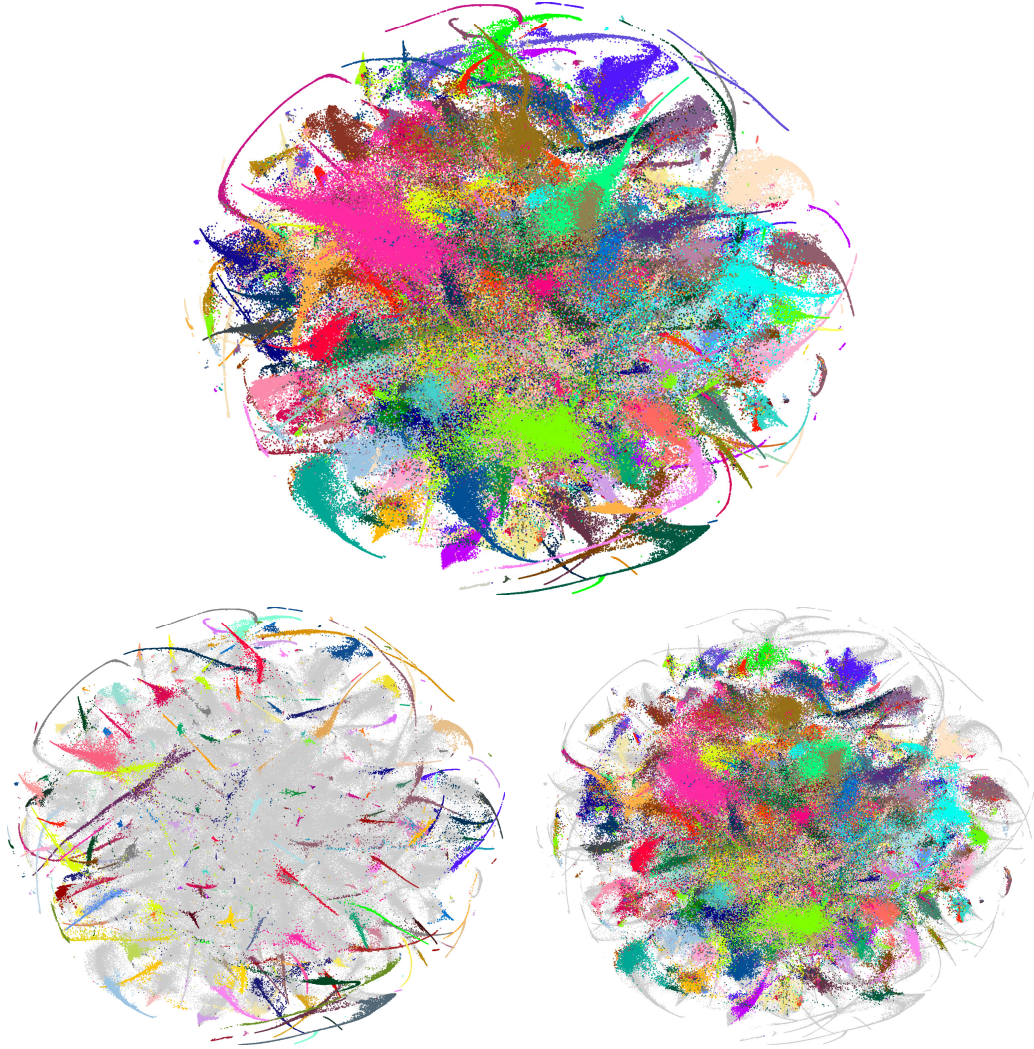
the 23 metatopics. We find that R-squares are roughly halved when moving from the 180 topics to the 23 metatopics. This comparison would not be interesting in OLS regressions. But because in both cases we select a Lasso model with exactly five regressors, the comparison is valid and suggests that the more granular 180-topic level of aggregation provides a superior summary of the state of the economy.

IA.F. Article-level News Attention

Time series plots describe the attention paid to each topic by the *WSJ* as a whole. In this section, we explore patterns in topic attention at the article level. Article-level attention is a 180-dimension vector listing the proportion attention that the article pays to each topic. While this is an enormous reduction in dimensionality relative to the term count representation, the fits amount to a large dataset in their own right: a matrix of roughly 800,000 articles by 180 topics. In order to visualize and interpret article-level fits, we project the fitted proportions into two dimensions using *t*-distributed stochastic nearest neighbor embedding (*t*-SNE, Maaten and Hinton, 2008). Intuitively, the *t*-SNE plot assigns a two-dimensional coordinate to each article so that the nearness of articles in the plane best represents the nearness of the article fits in the full 180-dimensional space.

The top panel of Figure IA.5 shows the two-dimensional embedding of model fits for every *WSJ* article in our data set, and Figure IA.6 breaks down the complexity of Figure IA.5 by separately plotting articles based on their leading topic for an illustrative subset of nine topics. In Figure IA.5,

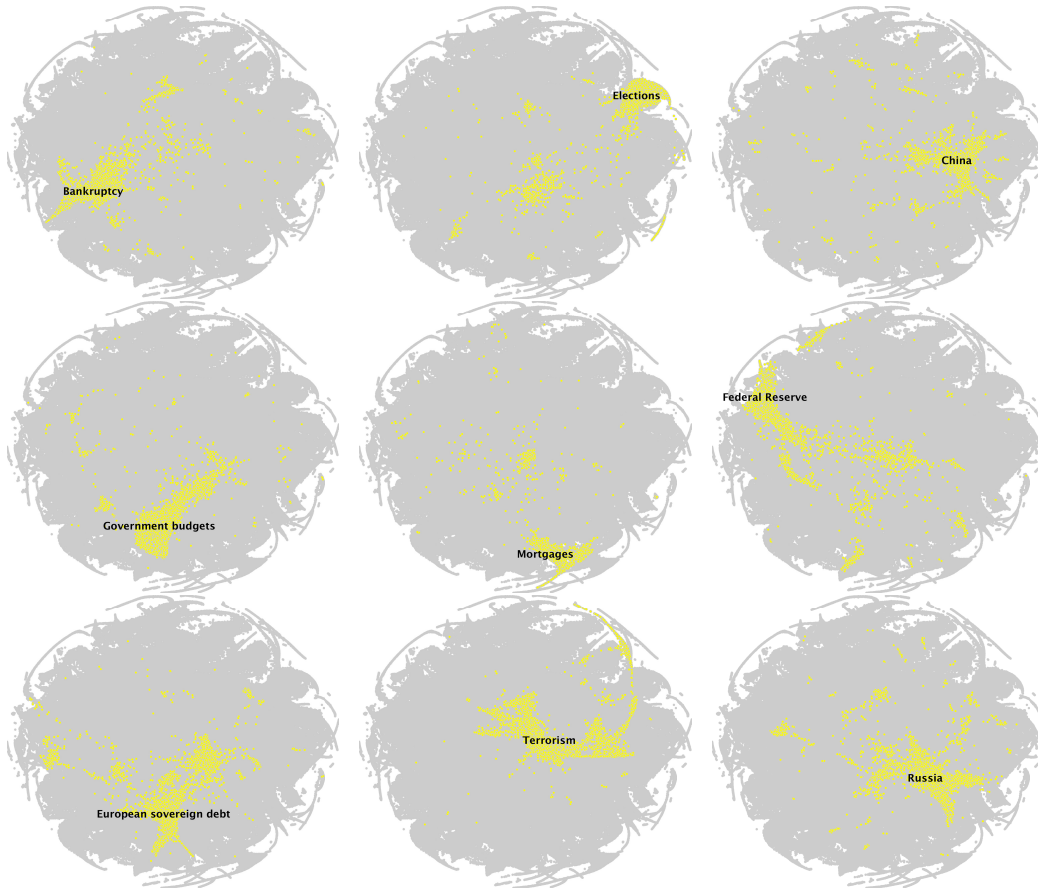
Figure IA.5. Article-level Nearest Neighbor Embedding



The top shows the two-dimensional embedding of *WSJ* articles, where each topic is assigned a color and articles are colored in correspondence with their dominant topic. The bottom left panel highlights only “pure” (those with maximum topic proportion exceeding 33%), while the bottom right shows “mixed” articles (maximum topic proportion below 25%).

each topic is assigned a color, and each article (a single point) is colored in correspondence with the topic having the highest weight in that article. Nat-

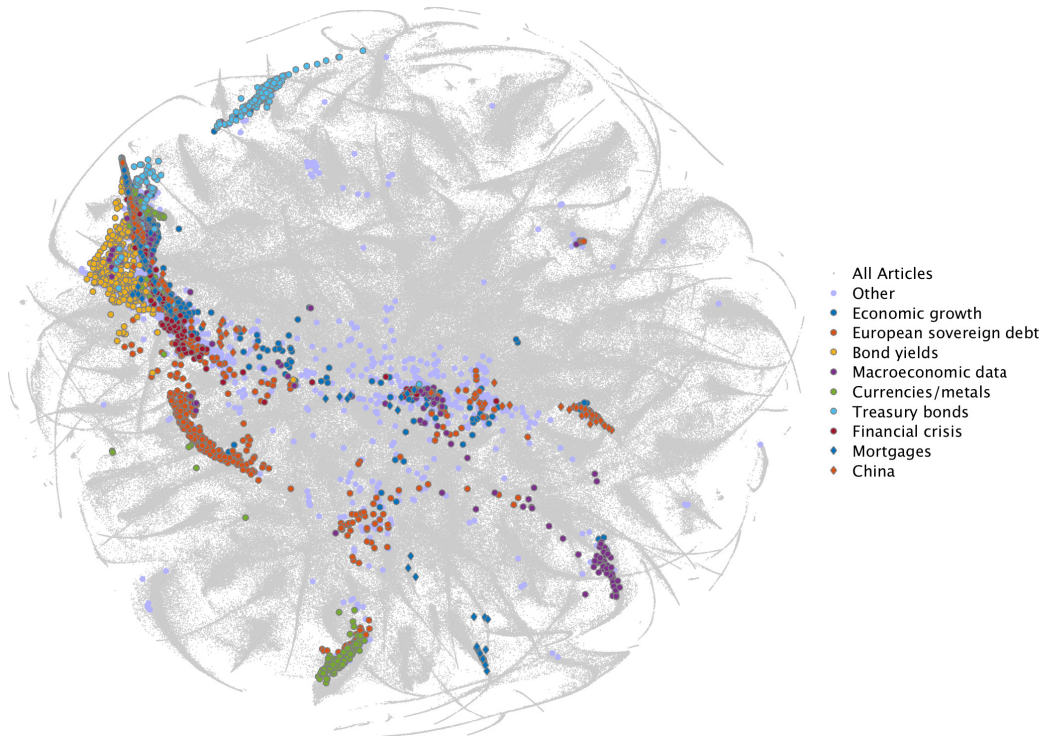
Figure IA.6. Article-level Nearest Neighbor Embedding (Topic Breakout)



Two-dimensional embedding of *WSJ* articles broken out by individual topics.

urally, articles cluster spatially based on their dominant topics. Graphically, an article (which the model views as a mixtures of topics) is an intersection of topic clouds. Articles that mix many topics must intersect with many clouds, so they are centrally located in the plot. Similarly, clouds of topics that tend to co-occur with other topics (such as “Federal Reserve”) need to be dispersed throughout the t -SNE, while those that mix with only a few other topics are more concentrated (such as “mortgages” and “terrorism”).

Figure IA.7. Articles Featuring the Federal Reserve



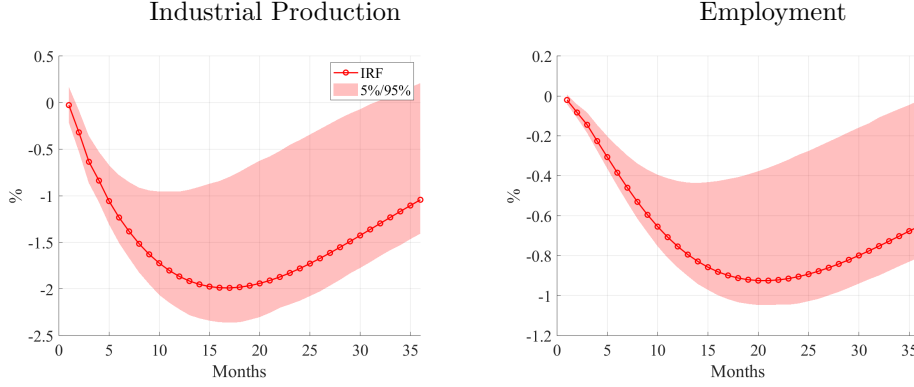
Detail of two-dimensional embedding for article whose dominant topic is “Federal Reserve.” Articles within this set are colored according to their second largest topic proportion.

This can also be seen at the article level via the two bottom panels of Figure [IA.5](#), which separates articles by their purity. “Pure” articles are shown in color on the left, and are defined as those with a maximum topic proportions exceeding 33%. These articles (18% by count) are dominated by a single topic. The right shows the set of “mixed” articles (maximum topic proportion below 25%). Pure articles, which have little need for proximity to other topics, and tend to live on the periphery of the embedding. Mixed articles require topic crossing and are thus centrally located.

Figure IA.7 provides more detail for one specific example. It documents the other topical content of articles whose leading topic is “Federal Reserve.” It splits articles within the Fed topic based on their second largest topic allocation. Figure IA.7 shows that articles about the Fed also tend to relate to the “economic growth,” “European sovereign debt,” “bond yields,” “macroeconomic data,” “Treasury bonds,” “financial crisis,” “mortgages,” and “China” topics, with each of these secondary topics tightly spatially clustered within the Fed topic.

Like the hierarchical metatopic analysis, the t -SNE plots as a whole demonstrate coherence of the estimated topic model. They show that an economic event—captured verbally as a news article—is summarized via our model as the constellation of topics that constitute that event. Likewise, the point-in-time distribution of aggregate topic proportions summarizes the overall state of the economy. These model estimates illustrate that the economy is fundamentally complex. An attempt at parsimoniously representing the verbal state of the economy, while a drastic simplification of the full *WSJ* narrative, nonetheless requires an elaborate specification to capture its common themes. But our model estimation results, and our further empirical analysis below, also demonstrate that this complexity can be effectively harnessed for economic analysis.

Figure IA.8. Impulse Responses With Confidence Intervals Adjusted for Lasso Selection



Responses of output and employment to a “recession” attention impulse in various VAR specifications, 1985–2017. Confidence intervals are adjusted to reflect lasso model selection.

IA.G. Post-selection Inference For Impulse Responses

When developing our text-augmented VAR, we use group-lasso to select news topics for model inclusion. This selection step affects inferences of statistical significance for our impulse response estimates. In this section we describe a post-selection inference method for constructing impulse response confidence intervals that account for group-lasso selection.

Our method to construct appropriate confidence intervals builds on the bootstrap procedure of Kilian (1999) that we use in Figure 4. First, we construct VAR residuals from our baseline VAR specification (which includes “recession” attention as the only news-based variable). Next, we resample residuals to produce bootstrap data samples. In each bootstrap sample, we select topics for inclusion in the VAR based on the cross-validated value

of $\lambda = 0.054$ from our analysis in Figure 9. All topics that receive non-zero group-lasso estimates based on this value of λ are included in the VAR model, which we re-estimate with OLS. From this we produce the “recession” attention impulse response function (if “recession” is not selected in the VAR for a given bootstrap sample, the impulse responses are set to zero). Confidence intervals are then defined as the point-wise 5% to 95% range of bootstrapped impulse response functions.

Figure IA.8 reports the post-selection adjusted confidence intervals for our baseline VAR specification. We note that the confidence intervals are somewhat wider than those in Figure 4, but that the quantitative magnitudes are similar and our conclusions regarding statistical significance of the impulse responses are unaffected by the post-selection adjustment.

IA.H. Macroeconomic VAR Impulse Responses Robustness

For brevity, in the paper we reported impulse responses visually in Section IV. Here, we supplement Figure 4 which only reports confidence intervals for a single alternative specification with confidence intervals for all specifications every five years.

Table IA.III reports point estimates and confidence intervals for industrial production and employment. The table shows for all alternative specifications we can reject a null effect for a considerable period of time following a “recession” attention impulse. Baseline reports our main specification results. oLDA reports the comparable results for an instance of our oLDA

specification and EPU reports the comparable results using EPU in place of our recession topic. News last places our recession topic last in the VAR specification, while news 2nd places it second. Incl. EPU, Mich, and VIX, each represent a specification that includes the corresponding variable. Incl. Trend includes a trend term and 6 Lags uses 6 lags instead of 3.

Table IA.III. Recession Impulse Response

Panel A: Industrial Production										
Specification	1	5	10	15	20	25	30	35		
Baseline	-0.03 [-0.16 0.12]	-1.03 [-1.27 -0.69]	-1.69 [-2.02 -1.03]	-1.93 [-2.30 -0.99]	-1.90 [-2.29 -0.77]	-1.69 [-2.09 -0.45]	-1.40 [-1.77 -0.09]	-1.08 [-1.50 0.21]		
oLDA	-0.01 [-0.17 0.15]	-0.84 [-1.16 -0.45]	-1.47 [-1.91 -0.74]	-1.77 [-2.23 -0.77]	-1.82 [-2.27 -0.61]	-1.71 [-2.11 -0.39]	-1.51 [-1.84 -0.14]	-1.28 [-1.56 0.09]		
EPU	-0.11 [-0.25 0.03]	-0.73 [-1.04 -0.33]	-0.96 [-1.37 -0.28]	-0.88 [-1.33 -0.06]	-0.68 [-1.15 0.19]	-0.45 [-0.95 0.41]	-0.22 [-0.76 0.60]	-0.03 [-0.59 0.72]		
News Last	0.00 [0.00 0.00]	-0.67 [-0.86 -0.42]	-1.08 [-1.38 -0.57]	-1.22 [-1.60 -0.50]	-1.18 [-1.61 -0.34]	-1.01 [-1.46 -0.11]	-0.78 [-1.23 0.13]	-0.53 [-0.98 0.32]		
News 2nd	-0.10 [-0.23 0.03]	-0.84 [-1.06 -0.53]	-1.26 [-1.59 -0.69]	-1.37 [-1.79 -0.57]	-1.27 [-1.75 -0.32]	-1.04 [-1.54 -0.00]	-0.73 [-1.25 0.30]	-0.42 [-0.96 0.54]		
Incl. EPU	-0.06 [-0.19 0.09]	-1.08 [-1.32 -0.73]	-1.68 [-2.00 -1.03]	-1.90 [-2.26 -0.96]	-1.87 [-2.24 -0.72]	-1.68 [-2.06 -0.44]	-1.41 [-1.79 -0.14]	-1.11 [-1.49 0.14]		
Incl. Mich	-0.02 [-0.16 0.11]	-0.99 [-1.23 -0.64]	-1.55 [-1.85 -0.91]	-1.70 [-2.03 -0.83]	-1.61 [-1.96 -0.59]	-1.40 [-1.75 -0.31]	-1.15 [-1.49 -0.05]	-0.91 [-1.27 0.16]		
Incl. VIX	0.01 [-0.13 0.16]	-1.08 [-1.32 -0.67]	-1.75 [-2.10 -0.97]	-1.99 [-2.38 -0.88]	-1.92 [-2.36 -0.58]	-1.67 [-2.10 -0.24]	-1.32 [-1.73 0.10]	-0.97 [-1.39 0.38]		
Incl. Trend	-0.02 [-0.15 0.11]	-1.02 [-1.25 -0.68]	-1.66 [-1.93 -0.98]	-1.87 [-2.16 -0.87]	-1.81 [-2.10 -0.59]	-1.59 [-1.87 -0.26]	-1.31 [-1.62 0.04]	-1.04 [-1.38 0.29]		
6 Lags	0.01 [-0.12 0.13]	-0.94 [-1.23 -0.55]	-1.51 [-1.92 -0.79]	-1.61 [-2.07 -0.63]	-1.47 [-1.98 -0.39]	-1.25 [-1.78 -0.16]	-0.99 [-1.51 0.06]	-0.74 [-1.26 0.27]		
Panel B: Employment										
	1	5	10	15	20	25	30	35		
Baseline	-0.02 [-0.04 0.00]	-0.30 [-0.36 -0.21]	-0.64 [-0.74 -0.42]	-0.84 [-0.97 -0.48]	-0.91 [-1.05 -0.43]	-0.87 [-1.03 -0.33]	-0.78 [-0.94 -0.19]	-0.66 [-0.83 -0.06]		
oLDA	-0.02 [-0.05 0.00]	-0.27 [-0.34 -0.16]	-0.58 [-0.71 -0.32]	-0.79 [-0.94 -0.37]	-0.89 [-1.05 -0.36]	-0.89 [-1.04 -0.28]	-0.84 [-0.96 -0.18]	-0.75 [-0.85 -0.09]		
EPU	-0.03 [-0.05 -0.00]	-0.17 [-0.26 -0.07]	-0.31 [-0.45 -0.08]	-0.33 [-0.52 -0.03]	-0.29 [-0.49 0.06]	-0.21 [-0.44 0.16]	-0.12 [-0.37 0.25]	-0.03 [-0.30 0.31]		
News Last	0.00 [0.00 0.00]	-0.15 [-0.19 -0.09]	-0.35 [-0.45 -0.19]	-0.47 [-0.61 -0.21]	-0.50 [-0.68 -0.17]	-0.47 [-0.67 -0.09]	-0.40 [-0.60 -0.00]	-0.30 [-0.51 0.08]		
News 2nd	-0.02 [-0.04 0.00]	-0.22 [-0.28 -0.13]	-0.44 [-0.55 -0.25]	-0.56 [-0.72 -0.25]	-0.58 [-0.77 -0.19]	-0.52 [-0.74 -0.08]	-0.42 [-0.65 0.05]	-0.29 [-0.55 0.16]		
Incl. EPU	-0.02 [-0.04 0.00]	-0.30 [-0.35 -0.20]	-0.63 [-0.72 -0.41]	-0.82 [-0.94 -0.47]	-0.89 [-1.02 -0.42]	-0.87 [-1.00 -0.31]	-0.79 [-0.93 -0.19]	-0.67 [-0.82 -0.07]		
Incl. Mich	-0.02 [-0.04 0.01]	-0.27 [-0.33 -0.18]	-0.57 [-0.66 -0.36]	-0.71 [-0.83 -0.39]	-0.74 [-0.87 -0.33]	-0.69 [-0.84 -0.22]	-0.60 [-0.76 -0.10]	-0.51 [-0.66 -0.01]		
Incl. VIX	-0.03 [-0.05 -0.00]	-0.31 [-0.37 -0.20]	-0.68 [-0.76 -0.41]	-0.88 [-0.99 -0.46]	-0.94 [-1.07 -0.38]	-0.89 [-1.02 -0.25]	-0.77 [-0.92 -0.10]	-0.64 [-0.78 0.03]		
Incl. Trend	-0.02 [-0.04 0.00]	-0.30 [-0.35 -0.20]	-0.64 [-0.72 -0.40]	-0.84 [-0.94 -0.45]	-0.91 [-1.00 -0.39]	-0.88 [-0.97 -0.28]	-0.79 [-0.88 -0.15]	-0.68 [-0.77 -0.02]		
6 Lags	-0.01 [-0.03 0.01]	-0.25 [-0.30 -0.16]	-0.53 [-0.63 -0.33]	-0.70 [-0.84 -0.38]	-0.77 [-0.94 -0.34]	-0.75 [-0.95 -0.26]	-0.67 [-0.89 -0.14]	-0.56 [-0.78 -0.03]		

Responses of output to a “recession” attention impulse in various VAR specifications, 1985–2017. Each column reports a given horizon for the impulse response function. The first number reports the point estimate, while the numbers in brackets report the lower and upper bounds of the 90% confidence interval.

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